SCHOOL OF COMPUTATION,
INFORMATION AND TECHNOLOGY —
INFORMATICS

TECHNICAL UNIVERSITY OF MUNICH

Bachelor's Thesis in Informatics

Explaining Visual Cluster Structure in High-Dimensional Data Through Hierarchical Decomposition and Predicate Induction

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Erklärung visueller Clusterstruktur in hochdimensionalen Daten durch hierarchische Zerlegung und Prädikatinduktion

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Software used

I used the following software when writing my thesis:

Used software	Details	
GitHub Copilot (VS Code)	**Use case:**	Python inline code completions
	Scope:	Implementation, source code
	Accuracy:	Only used for minor inline code completions
Claude Code and Claude agent sessions (Anthropic)	**Use case:**	Assistance in Implementation and debugging of the WebApp, Experiment harnesses (from author's designs), Very limited assistance for the python backend (calculations and actual analysis).
	Scope:	Implementation, source code
	Accuracy:	All changes reviewed by the author
Claude and ChatGPT	**Use case:**	Brainstorming and feedback on ideas and experiment design; language editing and stylistic refinement, including assistance with phrasing, grammar and clarity; adversarial review of thesis and code; Discovering relevant literature; Claude DataViz Skill for help with some visualizations;
	Scope:	Entire Work
	Comments:	Each finding and edit individually reviewed; research questions, experiment designs and all content decisions are the author's
	Accuracy:	Reported numbers cross-checked against the experiment artifacts; the author reviewed and is responsible for the final text
Zotero	**Use case:**	Reference and bibliography management
	Scope:	Entire work
	Comments:	Reference manager (no AI/generative use)
Google Scholar	**Use case:**	Literature search and discovery
	Scope:	Entire work

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Abstract

High-dimensional datasets are difficult to interpret directly and the two-dimensional embeddings used to explore them show *that* structure exists without explaining *which original features* produce it. A single global view also washes out structure that only exists in parts of the data. This thesis develops and evaluates a hierarchical, density-based approach to exploring high-dimensional tabular data. The dataset is recursively decomposed into density-defined regions, and each region is shown in its own, locally fitted two-dimensional projection, so the user can drill down from a global overview to compact regions and judge every view by the quality scores reported alongside it. Points the user selects within a region are then described in plain terms, as ranges over a few original features. Next to this exact description, a relaxed variant trims each range toward the bulk of the selection, trading some coverage for robustness to how exactly the points were selected. The approach is implemented as an interactive dashboard, Hierarchical Local Decomposition and Explanation (HiLDE), and evaluated in four experiments on synthetic and benchmark datasets. The experiments show that the approach helps where its assumptions hold: local views are more faithful than one projection fitted to the whole dataset when the projection method targets global structure, the hierarchy recovers nested cluster structure that a single flat clustering merges, and relaxed descriptions are shorter and less sensitive to the exact selection where clusters are well separated. They equally show where it does not: the local views do not improve for t-SNE, a single flat clustering recovers coarse labeled classes better, trimming the range of every feature at once over-tightens descriptions in high dimensions, and the proposed asymmetric trimming rule works no better than a simple even trim.

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1. Introduction

Real-world datasets are routinely high-dimensional. Each record is described by tens or hundreds of measurements. People cannot perceive more than three dimensions simultaneously, so analysts look for structure by reducing the data to two dimensions and viewing it as a scatter plot. Such dimensionality reduction (DR) is reasonable because high-dimensional data typically concentrates near lower-dimensional structure, and low-dimensional scatter plots have become a standard first step in exploring an unfamiliar dataset [Esp+21; Liu+17; NA19].

Such a picture, however, only tells the analyst *that* groups or patterns exist. It does not explain *why* a group is a group, that is, which of the original measurements make it distinct. Closing this gap between a visible pattern and a human-readable explanation is the first motivation of this work. The second is that interesting structure is highly subjective and often depends on only a handful of features at a time, and these features differ from one part of the data to another. A single overall projection therefore cannot do justice to every region of a varied dataset: sub-structure that matters locally is easily washed out in one global view, and a feature that carries such sub-structure can look entirely uninformative at the global scale (Figure 1.1).

Two linked difficulties follow. First, because what counts as *interesting* depends on the analyst and the task, the goal is not to *compute* interestingness but to provide a way to break the data into meaningful regions, view each one on its own terms, and give the analyst the characteristics and descriptions on which to base an informed judgment. The decomposition acts on *samples*: it carves the data into regions by density, so the feature subset relevant to a region is surfaced indirectly, through the region's locally adapted projection and the description that names its distinguishing features. Second, once an interesting region has been singled out, it should be described in plain terms, as simple ranges on a few original features, so the description can be read and trusted. A natural way to produce such a description, following the DimBridge system [Mon+25], is to turn a user's selection of points into feature ranges. The catch is that such a description can cling too tightly to the exact points chosen: change the selection a little and the description can change a lot. Here, we ask whether organizing the exploration as a hierarchy of local views, and *relaxing* the resulting descriptions, makes the process more reliable.

Interactive DR systems let the user steer a projection, yet they often operate on a

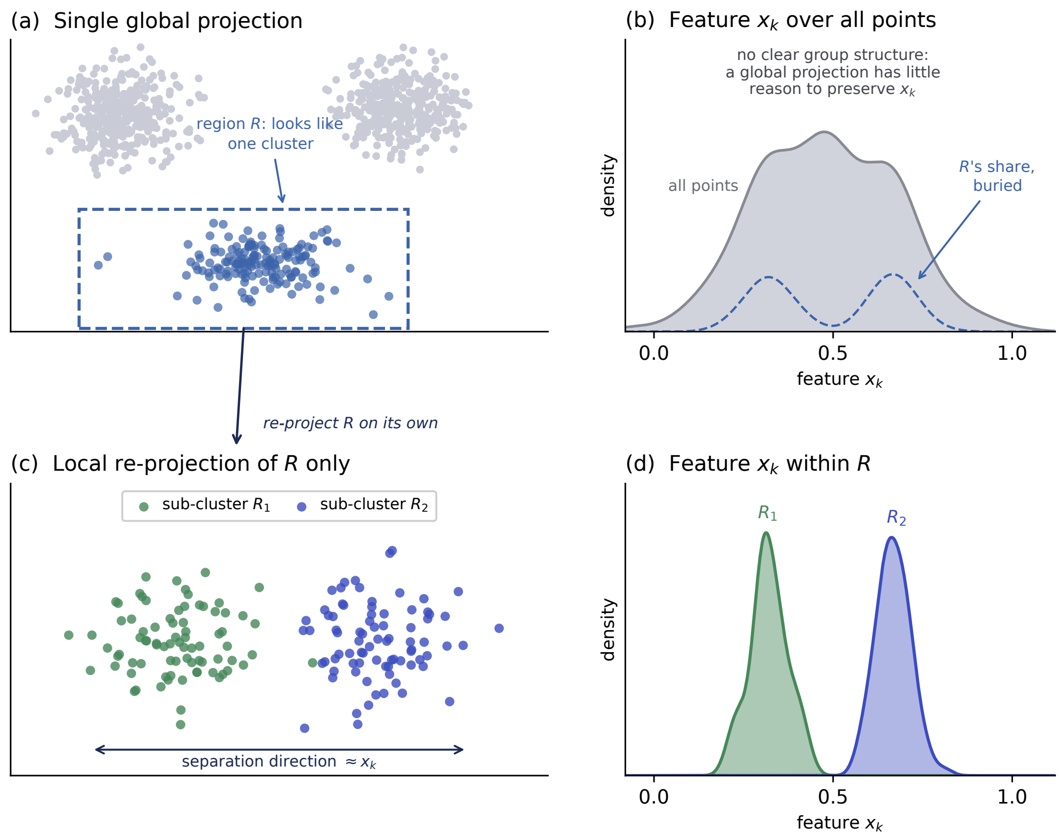


Figure 1.1.: (a) In a global projection, region R appears as one cluster, and (b) over the full dataset, feature x_k shows no clear group structure, the bimodal contribution of R's points (dashed) buried among the rest. (c) Re-projecting only R separates two sub-clusters, and (d) within R the values of x_k alone tell them apart. *(Illustrative synthetic data; both projections schematic.)*

single *global* view, and the hierarchical exceptions either refine one precomputed global structure or re-project user-chosen subsets, without describing a drilled-into region by feature ranges. Pattern-explanation methods such as DimBridge [Mon+25] connect a visual selection back to the original features, but whether an explanation survives a perturbation of a single selection is not evaluated, even where consecutive selections are coupled by a smoothness term. Subspace-discovery methods automate the search for locally relevant feature subsets, but do not combine that search with an interactive, interpretable description. And the *stability* of an explanation (whether it survives a slightly different selection or a recomputed projection) is rarely evaluated at all. In short, the current body of related work does not provide the combination of interactive drill-down into locally re-projected regions, interpretable feature-range descriptions of a selected point set, and an evaluation of how stable those descriptions are.

Contributions. The contributions of this thesis are as follows:

1. A hierarchical, density-based exploration framework that recursively decomposes a dataset into regions, each summarized by its own two-dimensional projection, cluster characteristics, and outlier scores.
2. A predicate-description layer offering a strict, DimBridge-style range predicate and a relaxed variant that trims each feature range asymmetrically toward the selection's center mass.
3. An interactive prototype, HiLDE (Hierarchical Local Decomposition and Explanation), integrating drill-down, selection, and explanation in a single workflow.
4. An evaluation protocol combining projection-faithfulness and predicate-level measures with a scripted stability harness, applied in four experiments on synthetic and benchmark data.

Outline. Chapter 2 introduces the concepts the work builds on: high-dimensional data and subspaces, dimensionality reduction and how its quality is measured, density-based clustering, and predicates as interpretable descriptions. Chapter 3 positions the work against interactive DR, pattern-explanation, predicate- and rule-based, and subspace-discovery methods. Chapter 4 describes the approach: the recursive density-based hierarchy, the strict and relaxed predicate methods, and the configuration and evaluation design built around them. Chapter 5 documents the HiLDE prototype. Chapter 6 states four evaluation questions, one per experiment, and reports the experiments that answer them. Chapter 7 interprets the findings and bounds the conclusions, and Chapter 8 summarizes the work, answers the evaluation questions directly, and outlines future work.

2. Background

2.1. High-Dimensional Data, Manifolds, and Subspaces

As the number of features d describing each record grows, geometry becomes counter-intuitive. First, norms and pairwise distances *concentrate* (Figure 2.1a): for points with d independent standard-normal features, distances concentrate around their mean with a spread that does not grow with d, so nearly all points lie in a thin shell and the contrast between a point's nearest and farthest neighbor shrinks toward zero [AHK01; Bey+99]. Second, volume behaves strangely (Figure 2.1b): the volume of the unit ball, and even the surface area of the unit sphere, approaches zero as d grows, while the volume of the enclosing cube $[-1, 1]^d$ grows as 2^d, so a sample of fixed size becomes exponentially sparse. These effects, collectively called the *curse of dimensionality*, weaken any notion of similarity computed over all features at once [AHK01; Bey+99; KMB12; Mül+09].

Dimensionality reduction is nonetheless reasonable because of the *manifold assumption*: although data is recorded in a high-dimensional space, it typically varies along far fewer intrinsic degrees of freedom and approximately lies on a lower-dimensional *manifold*, a subset that locally resembles a Euclidean space of much smaller dimension, the way the surface of a ball in three-dimensional space is locally two-dimensional. The assumption is plausible in practice because recorded features are rarely independent. They are generated by fewer underlying factors and tied together by physics, redundant sensors, or the measurement process. Images of one object under a moving camera, for example, are recorded as thousands of pixel features but vary along a handful of pose and lighting parameters. In principle the data can therefore be re-expressed in a space of the manifold's intrinsic dimension with little loss, which is what DR attempts [Esp+21; NA19].

A complementary concept is that of a subspace. Given a dataset with feature set $F = \{f_1, \ldots, f_d\}$, a *subspace* here is an *axis-aligned* one: a subset $S \subseteq F$ of the original features, so that restricting the data to S is an axis-parallel projection onto the $|S|$-dimensional coordinate space. (Arbitrary linear combinations of features, as in the linear-algebra sense of the word, are not considered.) A region of interest is then a set of points together with a subspace in which they are tightly grouped or otherwise exceptional, while the remaining features $F \setminus S$ behave like noise for that region. The working assumption of the subspace literature, and of this work, can be

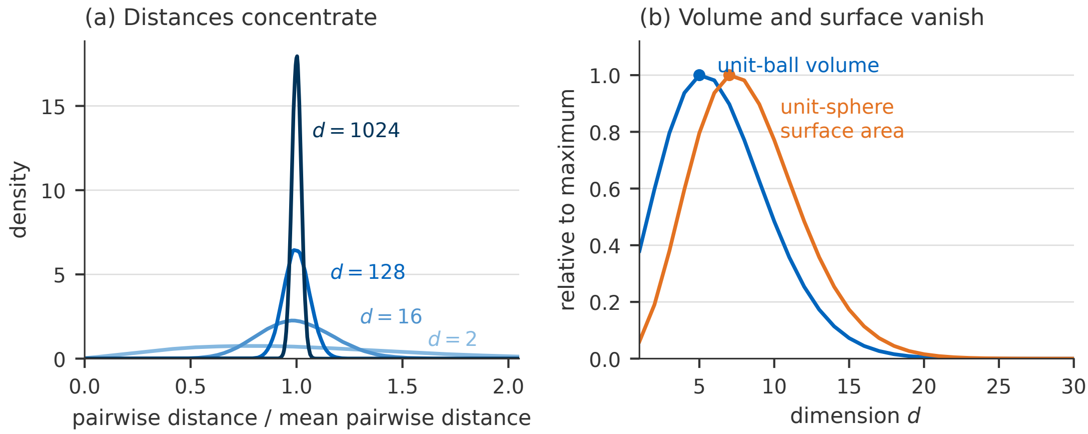


Figure 2.1.: Two effects of the curse of dimensionality. (a) Pairwise distances between 500 points with d independent standard-normal features, rescaled by their mean: as d grows the distribution tightens, and the median contrast between a point's farthest and nearest neighbor, $(\delta_{\max} - \delta_{\min})/\delta_{\min}$, falls from $\approx$53 at $d = 2$ to 0.12 at $d = 1024$. (b) Volume of the unit ball and surface area of the unit sphere, each relative to its maximum (5.26 at $d = 5$ and 33.1 at $d = 7$): beyond these peaks both fall toward zero.

read as an axis-aligned counterpart of the manifold assumption: when a dataset pools measurements of heterogeneous processes, each group of points is often governed by a small subset of the recorded features while the rest carry no signal for that group [KMB12; Mül+09]. This gives the introduction's informal notion of locally relevant structure a concrete form: interesting structure is often *subspace-local*. It underlies both the locally adaptive hierarchy (different regions are interesting in different subspaces) and the range predicates (a description names the few features of S and their ranges).

2.2. Dimensionality Reduction

Dimensionality reduction maps high-dimensional points into a lower-dimensional space, while trying to preserve some aspect of the original structure [Esp+21; NA19]. Techniques are commonly split into linear and non-linear groups. Principal Component Analysis (PCA) is linear: it projects onto the orthogonal directions of maximal variance. Multidimensional scaling (MDS) seeks a low-dimensional layout that preserves pairwise dissimilarities and its variants span both families. *Classical* MDS [Tor52] is linear: it recovers coordinates from an eigendecomposition of the double-centered distance matrix and, with Euclidean distances, is equivalent to PCA. *Metric* MDS is a different method: it minimizes a stress objective over the embedding coordinates directly [BG05; Lee77], and is non-linear, as is Kruskal's *non-metric* MDS, which preserves only the rank order of the dissimilarities [Kru64]. MDS in this work always means metric MDS. Manifold methods such as t-distributed stochastic neighbor embedding (t-SNE) [VH08] and Uniform Manifold Approximation and Projection (UMAP) [MHM18a] are non-linear and optimize for preserving local neighborhoods. They typically produce visually well-separated clusters, at the cost of distorted global distances and stochastic, parameter-sensitive output, documented in detail for t-SNE [WVJ16].

The central trade-off is between preserving *local* and *global* structure, and no projection preserves both perfectly [Esp+21; NA19]. The developed HiLDE Dashboard prototype (Chapter 5) exposes PCA, t-SNE, MDS, and UMAP. UMAP also serves as a pre-clustering reduction to a configurable number of components, on which density-based clustering is more tractable, though the documentation cautions that UMAP does not completely preserve density and, like t-SNE, can create false tears in clusters, yielding a finer clustering than is present in the data [MHM18b].

2.3. Measuring Reduction Quality

Because every projection is lossy, an embedding must be assessed for how faithfully it preserves the original structure [Aup07; LV09]. The measures follow the ZADU library [Jeo+23], which organizes distortion measures into local, cluster-level, and global families. Only the measures used are described here.

Throughout, let n points have pairwise distances d_{ij} in the original space and $\hat{d}_{ij}$ in the embedding. For a neighborhood size k, let $N_k(i)$ and $\hat{N}_k(i)$ be the k nearest neighbors of point i in the original and the embedded space, and let $r(i,j)$ and $\hat{r}(i,j)$ be the rank of j by distance from i in each space (rank 1 is the nearest neighbor).

At the *local* level, *trustworthiness* and *continuity* quantify how well k-nearest-neighbor relations survive the projection [LV09; VK01]. Trustworthiness charges the embedding for the spurious neighbors it introduces,

$$T(k) = 1 - \frac{2}{nk(2n-3k-1)} \sum_{i=1}^{n} \sum_{j \in \hat{N}_k(i) \setminus N_k(i)} \big(r(i,j) - k\big), \tag{2.1}$$

and continuity $C(k)$ is defined symmetrically with the two spaces exchanged: it sums $\hat{r}(i,j) - k$ over the original-space neighbors $j \in N_k(i) \setminus \hat{N}_k(i)$ that the embedding pulls apart. Both lie in $[0,1]$, higher better. The *mean relative rank errors* (MRRE) refine this by weighting every neighbor by how far its rank has moved [LV09]:

$$\mathrm{MRRE}_{\text{false}}(k) = \frac{1}{n\,H_k} \sum_{i=1}^{n} \sum_{j \in \hat{N}_k(i)} \frac{|r(i,j) - \hat{r}(i,j)|}{\hat{r}(i,j)}, \qquad H_k = \sum_{l=1}^{k} \frac{|n-2l+1|}{l}, \tag{2.2}$$

with the companion term $\mathrm{MRRE}_{\mathrm{missing}}$ summing $|r(i,j) - \hat{r}(i,j)| \, / \, r(i,j)$ over $j \in N_k(i)$ instead. Both are reported in inverted form ($1 - \mathrm{MRRE}$, higher better). At the *global* level, *stress* measures the overall mismatch between original and embedded pairwise distances [Kru64], here in the normalized form

$$\sigma = \sqrt{\sum_{i<j} \left(d_{ij} - \hat{d}_{ij}\right)^2 \Big/ \sum_{i<j} d_{ij}^2}\,, \tag{2.3}$$

which is 0 for perfect distance preservation, lower better. At the *cluster* level, Steadiness and Cohesiveness measure whether clusters separated in the original space remain separated, and undispersed, in the projection [Jeo+22]. The prototype instead reports the class angular distortion index (CADI), which measures how the angular arrangement of labeled groups is distorted [GKM26] and takes the child-cluster labels the tree supplies naturally. Reported at internal nodes with each node's child-cluster assignment as labels, CADI measures preservation of the identified child-cluster arrangement rather than agreement with external classes. Its formal definition is given in [GKM26] and is not restated here. Label-based evaluation must be treated with care, however, because classes are not clusters: points sharing a label need not form a single dense region [Jeo+24]. The hierarchy therefore treats density-based structure, not labels, as its primary signal.

Separately from these projection-faithfulness measures, the quality of a *clustering* can be assessed with an internal validity index. Density-Based Clustering Validation (DBCV) is designed for the density-based clusters this work produces: it scores a partition by the contrast between the least-dense within-cluster connection and the densest between-cluster separation [Mou+14]. Formally, each cluster C_i receives the validity

$$V(C_i) = \frac{\min_{j \neq i} \mathrm{DSPC}(C_i, C_j) - \mathrm{DSC}(C_i)}{\max\left\{ \min_{j \neq i} \mathrm{DSPC}(C_i, C_j),\ \mathrm{DSC}(C_i) \right\}}, \tag{2.4}$$

where the density sparseness $\mathrm{DSC}(C_i)$ is the heaviest edge of the cluster's internal minimum spanning tree under density-based reachability distances and the density separation $\mathrm{DSPC}(C_i, C_j)$ is the smallest such distance between the two clusters. DBCV is the size-weighted mean $\sum_i (|C_i|/n)\, V(C_i) \in [-1, 1]$, higher better [Mou+14]. It is the clustering-quality objective of the configuration search (Section 4.6).

2.4. Density-Based Clustering

A clustering algorithm takes a set of points and a dissimilarity measure and, without any inherent labels, partitions the points into groups (*clusters*) such that points within a group are more similar to one another than to points of other groups. The families of algorithms differ in how they approach this informal goal. Density-based clustering defines clusters as connected regions of high point density separated by sparser regions, which lets it recover arbitrarily shaped clusters and label low-density points as noise or outliers. Centroid- or model-based methods, by contrast, impose convex or Gaussian cluster shapes. Density-Based Spatial Clustering of Applications with Noise (DBSCAN) introduced this idea with a fixed density threshold [Est+96]. Hierarchical Density-Based Spatial Clustering of Applications with Noise (HDBSCAN) generalizes it across all density levels: it converts pairwise distances to mutual reachability distances, condenses the cluster hierarchy grown from their minimum spanning tree, and keeps the clusters that persist over the widest range of density levels [Cam+15; CMS13]. An accelerated implementation makes this fast enough for interactive use [MH17; MHA17].

HDBSCAN suits the hierarchical branch; it requires no fixed number of clusters, tolerates clusters of varying density and non-convex shape, and natively marks noise (label -1) [Cam+15]. Each HDBSCAN call uses its condensed tree to extract one flat set of clusters, which we use to recompute the analysis for each region (Section 4.3). As a by-product it yields Global-Local Outlier Score from Hierarchies (GLOSH) outlier scores, a global-local measure of how isolated each point is relative to the densest nearby cluster, which the prototype surfaces per layer [Cam+15].

2.5. Predicates as Interpretable Descriptions

Finding an interesting group is only half the task. It must also be described in a way a human can read. Following DimBridge [Mon+25], a region is described by a *predicate*: a conjunction of axis-parallel range constraints over original features, for example $x_i \in [a, b] \wedge x_j \in [c, d]$. Each clause names one feature and a range, so the predicate is interpretable by construction. Its *length* (the number of clauses) is a proxy for description complexity. *Relaxing* a predicate here means trimming each clause to a central quantile range of the selected values, so the description depends on the selection's bulk rather than its extremes.

3. Related Work

3.1. Interactive Dimensionality Reduction and Visual Analytics

In visual analytics, dimensionality reduction is regularly part of an interactive loop in which an analyst projects data, inspects the result, and adjusts the view. Sacha et al. provide a structured analysis of this human-in-the-loop process, cataloging the ways users steer DR and interpret its output [Sac+17]. Nonato and Aupetit survey multidimensional projection for visual analytics, linking projection techniques to the distortions they introduce and the analysis tasks they support [NA19], and Sedlmair et al. give empirical guidance on when scatter plots and particular DR techniques are appropriate [SMT13]. Broader surveys document a decade of advances in high-dimensional data visualization and a quantitative comparison of DR techniques [Esp+21; Liu+17]. This body of work often assumes a *single* projection that the user steers. A hierarchical strand is the exception. Hierarchical Stochastic Neighbor Embedding (HSNE) builds a hierarchy of embeddings over successively aggregated landmarks of one neighborhood graph, so that the analyst can drill from an overview embedding into more detailed ones [Pez+16], an interaction model applied at scale to single-cell data with millions of records [Une+17]. Its levels refine one precomputed global similarity structure. HUMAP brings hierarchical drill-down to UMAP, projecting user-chosen subsets while preserving the analyst's mental map across levels [Mar+25], and ExplorerTree supports multilevel focus-and-context exploration of a two-dimensional embedding [Mar+21]. None of these induces a feature-range description of a drilled-into region: the per-cluster marker statistics of HSNE's application are summaries, not induced range predicates. Our work recomputes the analysis (reduction, clustering, and projection alike) per region and complements each region with an interpretable description; the result is a *hierarchy* of local projections, one per region, so different parts of a heterogeneous dataset are each viewed through an adapted embedding.

3.2. Explaining Visual Patterns in Dimensionality Reductions

A precursor is the line of work that connects a visual selection in a DR plot back to an explanation in the original feature space. The most closely related system is

DimBridge: it explains a user-selected visual pattern with first-order predicates, conjunctions of feature-range constraints, built by recursive predicate induction: base clauses are expanded and re-scored until the score stops improving, retaining multiple overlapping candidate predicates. DimBridge adopts this algorithm from PIXAL as its accuracy-oriented option, re-scored with F1 [Mon+22; Mon+25]. Related explanation approaches differ in the form of the explanation rather than the goal: DimVis interprets selected clusters with an explainable boosting machine, reporting feature contributions [Sal+24]; DT-SNE structures a t-SNE visualization as a decision tree so that regions correspond to interpretable splits [BDF23]; and PIXAL reasons about anomalies over visual patterns [Mon+22]. Attribute-based explanation predates this line: da Silva et al. color a projection by the attributes that vary *least* over each point's neighborhood, explaining local similarity [Sil+15], and ccPCA contrasts a chosen cluster against the rest to find the features that set it apart [FKM20]. Both explain regions rather than user selections. InfoClus attaches sparse feature explanations to a partition of an embedding constrained to a hierarchical-clustering dendrogram, optimizing an explicit informativeness-versus-complexity trade-off [Lai+25]: hierarchical decomposition coupled to feature explanations, though the partition is computed rather than selected and the embedding is not recomputed per region. What these share, and what our work targets, is that they operate on *one* projection. DimBridge couples consecutive brushes with a smoothness term that keeps predicates consistent along a drawn trajectory. However, no reviewed system evaluates whether an explanation survives a perturbation of a single selection. This work makes that sensitivity visible by displaying the strict predicate beside its relaxed counterpart and quantifies it in the evaluation (Chapter 6).

3.3. Predicate- and Rule-Based Explanation Methods

Beyond the DR setting, many methods describe a group of points through interpretable rules or predicates over the original features. Decision-tree surrogates and rule-extraction methods produce conjunctive conditions that predict membership of a target group, and example-driven systems such as Seer generate extraction rules from user-specified instances [Han+17]. Subgroup-discovery descriptions likewise take the form of interpretable feature conditions [Her+11]. Outside the projection setting, Anchors learns rule-based explanations of the same form (conjunctions of feature predicates scored by precision and coverage) for the region where a model's prediction is locally stable [RSG18]. It explains a model rather than a user-selected point set, but its vocabulary is the one the predicate-quality measures here use. The predicate used here is a deliberately simple instance of this family, a per-feature range conjunction (Chapter 5). Its distinctive element is the relaxation operator: trimming each clause inward by an amount that depends on the skew of the selected values, an operation aimed at stability for which the methods reviewed here offer no direct analogue at the level of individual interval bounds.

3.4. Subspace Discovery and Relevant-Subspace Mining

A parallel line of work searches automatically for subsets of dimensions in which a subset of points is interesting. Subgroup discovery and exceptional model mining search for subsets that deviate from the global pattern with respect to a target or a model [Her+11; LFK08]. In the unsupervised setting, relevant-subspace clustering mines the most interesting non-redundant concepts in high-dimensional data [Mül+09], HiCS ranks high-contrast subspaces to support density-based outlier detection [KMB12], projection pursuit searches for informative low-dimensional projections [Hub85], and automated data slicing finds under-performing slices for model validation [Chu+20]. Apart from projection pursuit, which optimizes over continuous projection directions rather than discrete feature subsets, these methods perform an exhaustive or heuristic *combinatorial* search over subspaces. The combinatorial tradition goes back to CLIQUE [Agr+98] and is surveyed, with its successors, by Kriegel et al. [KKZ09]. Two neighboring families come closer to the route taken here and are worth distinguishing. Hierarchical subspace clustering discovers hierarchies *of subspace clusters* automatically (HiSC orders points by subspace-distance reachability [Ach+06], and its successor detects and visualizes the resulting cluster hierarchies [Ach+07]), but the hierarchy is mined, not steered: there is no interactive selection, no per-region re-projection, and no induced description whose stability could be evaluated. Interactive visual subspace exploration puts the analyst in the loop: SubVIS lets experts inspect patient subgroups across subspace clusterings [Hun+16], dynamic projections animate transitions between subspace views [Liu+15], and the Subspace Voyager navigates a continuum of 3D subspace projections [WM18]. But these systems explore precomputed or ball-shaped subspace structures of the *whole* dataset. None recomputes the analysis for a user-selected region, and none describes a region by induced feature ranges. The present work reaches candidate regions through an *interactive density-based hierarchy* and then describes them, trading the completeness of automated search for interpretability and user control.

3.5. The Gap Addressed by This Thesis

The four strands reviewed above each supply one piece and lack another. Table 3.1 summarizes the pattern, one judgment per section-closing contrast. No reviewed approach, including the hierarchical and visual subspace systems above, combines

Table 3.1.: Positioning of this thesis against the four strands of related work, by the four properties whose combination the reviewed literature does not provide. ✓ = provided, (✓) = partially or in some systems, ✗ = not provided, n/a = outside the strand's setting. HSNE's large-scale single-cell application reports per-cluster marker summaries, which are not induced range predicates, hence the ✗ in the first row's description column.

	Per-region re-projection	Interactive selection	Interpretable feature-range description	Stability evaluated
Interactive DR & VA	(✓)	✓	✗	✗
Pattern explanation in DR	✗	✓	✓	✗
Predicate- and rule-based methods	n/a	(✓)	✓	✗
Subspace discovery	✗	(✓)	(✓)	✗
This thesis	✓	✓	✓	✓

interactive selection, sample-conditioned per-region re-projection, and an interpretable feature-range description. InfoClus comes closest on the description side (Section 3.2), but its partition is neither interactively selected nor re-projected per region. And, among the methods reviewed here, the *stability* of an explanation under changes to the selection or the projection is not evaluated. Outside the projection setting, robustness of explanations has been examined for attribution methods [AJ18].

The contributions listed in Chapter 1 target these four deficits directly: the recursive, density-based hierarchical decomposition addresses the first and, by reaching candidate regions interactively, the subspace-discovery coupling. The relaxed predicates address the selection-sensitivity of existing pattern explanations, and the evaluation protocol addresses the fourth, including a stability harness that perturbs selections programmatically and compares the admitted sets across both predicate methods. How each is tested is stated as the evaluation questions (EQ1–EQ4) at the start of Chapter 6.

4. Methodology

4.1. Aims of the Approach

The gap of Section 3.5 translates into three aims the approach must meet. First, and primarily, it must surface and isolate locally relevant structure that a single global ("flat") projection washes out, by decomposing the data recursively with a density-based clustering and giving each region its own local dimensionality reduction. Second, it must describe a user-selected set of points within a region in interpretable terms (axis-parallel feature ranges) in a way that is robust to the exact composition of the selection, which motivates relaxing the strict, DimBridge-style predicate [Mon+25]. Third, every level of the process must be measurable, combining projection faithfulness per node (trustworthiness and continuity, MRRE, stress, class angular distortion, see Section 4.6) with predicate-level measures (precision, recall, coverage, predicate length), so that the analyst can tell a trustworthy view from a distorted one and the first two aims can be tested quantitatively.

Each subsequent section of this chapter traces back to one of these aims. How well the realized approach meets them is stated as the evaluation questions EQ1–EQ4 (Section 6.1) and tested by one experiment per question.

4.2. Overview of the Proposed Approach

The approach is a two-phase pipeline: a global, recursive density-based decomposition turns a flat dataset into a navigable hierarchy of regions. A local, interactive description phase then characterizes a selected set of points by range predicates.

The data flows as follows. A tabular dataset is read and the selected feature columns are prepared as described in Section 4.4. Then, a recursive HDBSCAN procedure decomposes this representation into a tree of regions. At each node, cluster characteristics, predicates, a local two-dimensional cluster projection, and projection-faithfulness scores form an overview from which the analyst drills down (Figure 4.1 shows this overview as the dashboard presents it). At a chosen leaf, a local two-dimensional projection is shown, the analyst selects a set of points, and that selection is described by feature distributions and a predicate in both its strict and relaxed form. Hierarchy construction

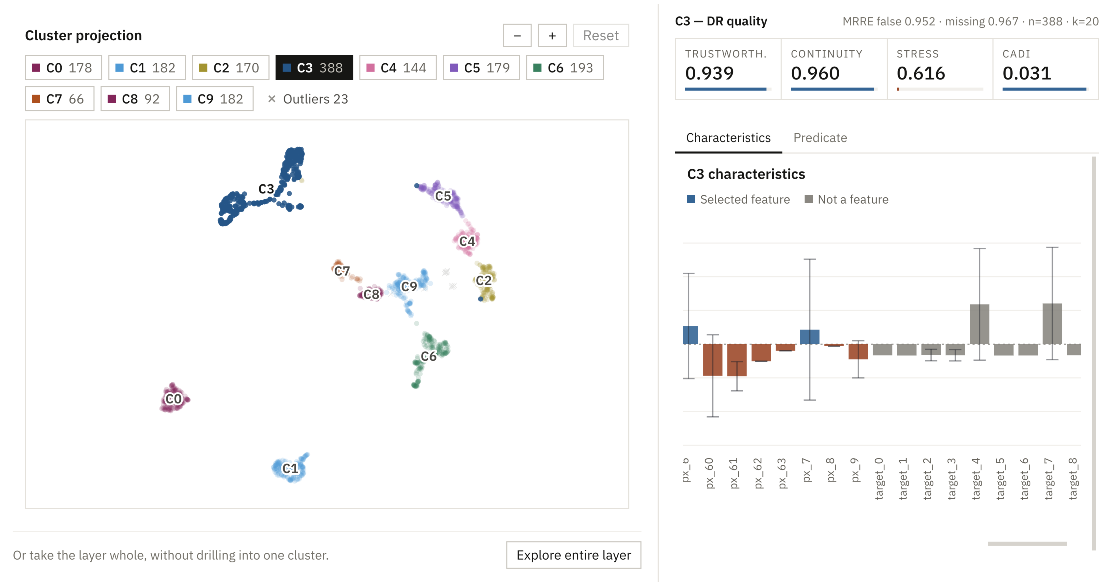


Figure 4.1.: The HiLDE dashboard: the cluster-projection overview used for drill-down, the per-node faithfulness scores, and the cluster characteristics of the selected cluster; the predicate view sits behind the adjacent tab. Example from the digits dataset [AK98].

and navigation serve the first aim. The predicate description of a selection serves the second, and the faithfulness and predicate scoring that runs across both phases serves the third.

4.3. Hierarchical Exploration Process

The dataset is explored level by level: each node is clustered and the procedure recurses on each resulting cluster, so the output is a tree whose leaves are compact, density-defined regions.

A node becomes a *leaf* when any of three stopping conditions holds: the depth limit is reached (depth $\geq$ `hierarchical_layers`); the node is too small to cluster meaningfully ($|X| < 2 \cdot$ `hclust_min_cluster_size`); or clustering yields fewer than two valid, non-noise clusters, leaving nothing to split. Otherwise the node is an *internal* node (Figure 4.2): each cluster HDBSCAN finds becomes a child, on whose points the build routine recurses. Points labeled as noise (-1) are excluded from every child, though they remain part of their parent node. The decomposition is thus sample-adaptive rather than subspace-adaptive: recursion conditions the analysis on ever smaller subsets

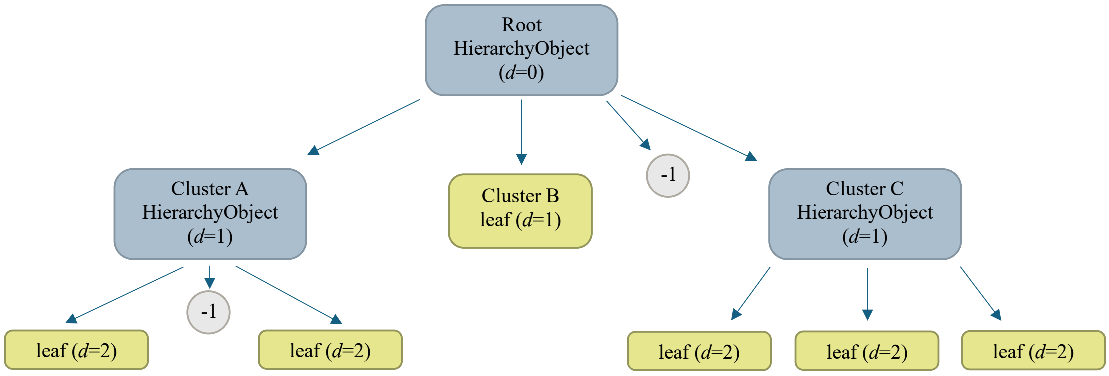


Figure 4.2.: Recursive decomposition of the original dataset. Internal nodes (blue) split further. Recursion stops, yielding leaves (green).

of rows, while the feature set is never selected or weighted per node. A region's relevant subspace is surfaced only indirectly, through its locally adapted projection and the predicate that names its distinguishing features. The planted-subspace experiment quantifies what this indirection costs against a subspace-aware oracle (Section 6.7).

Keeping the original row indices allows a selection made several levels deep, and the predicate that describes it, to be expressed in terms of the original records and features rather than an intermediate embedding. A leaf, together with the points selected within it, is therefore the unit of analysis for the leaf-level evaluation of Chapter 6 and the predicate experiment (EQ4). The contrast that motivates the hierarchy, and that EQ2 and EQ3 test, is this tree of locally adapted views against a single flat projection of the whole dataset: a flat projection must compromise across every region at once, whereas the hierarchy gives each region its own embedding.

4.4. Initial Dimensionality Reduction

Before clustering, the data is optionally standardized and optionally pre-reduced with UMAP [MHM18a] to a moderate number of components, so that HDBSCAN operates in a space of moderate dimensionality where density-based clustering remains practical [MHM18b].

Standardization, when enabled, z-scores the features with statistics fit once on the root node and reused at every level, keeping every node and score on one common scale, so faithfulness measures and the global predicate scope remain comparable across the tree rather than drifting as the data is subdivided. A local-scope predicate is instead standardized within its node, by design. Scaling matters for density-based clustering in particular: HDBSCAN's notion of density rests on distances, and an unscaled feature with a large numeric range would otherwise dominate them and mask structure in the remaining features.

The optional UMAP pre-reduction maps the data to `hclust_umap_n_components` dimensions, governed by the usual UMAP hyperparameters [MHM18a]. Like the standardization above, it is fit once on the root: where it is active, the recursion clusters ever smaller slices of that one embedding rather than re-reducing each node. Pre-reduction is itself a lossy transformation, however, so a cluster found in UMAP space need not correspond to a genuine cluster in the original feature space. This tension is what the per-node faithfulness scores surface. It is revisited in Chapter 7.

4.5. Cluster Identification

At every node, HDBSCAN [Cam+15; MHA17] identifies candidate regions as density-connected clusters, automatically determining their number and marking low-density points as noise.

Density-based clustering is a natural fit here (Section 2.4), and the recursion adds an argument of its own: because the method re-clusters at every level, any fixed cluster number k would be arbitrary, differing from one region to the next. HDBSCAN sidesteps both problems. Two hyperparameters govern its behavior: `hclust_min_cluster_size`, the smallest group that counts as a cluster, and `hclust_min_samples`, which controls how conservative the noise labeling is. Noise points (label -1) are handled as described in Section 4.3. Alongside the cluster labels, HDBSCAN yields the GLOSH outlier scores of Section 2.4, which the prototype surfaces per level.

Each internal node is summarized for the overview in three complementary ways. Per-cluster *characteristics* are the cluster's mean z-scores on the scale fit once at the root (Section 4.4), describing which features set the cluster apart from the dataset and in which direction (Figure 4.4). Numeric columns not selected as features (in the shipped datasets, the withheld target columns) are instead contrasted with the current node. The chart therefore mixes root-relative feature bars with node-relative non-feature bars: root scaling keeps feature bars comparable across the tree at the cost of a purely local contrast. The node's points are shown in its own two-dimensional *cluster projection*, with each child cluster colored, noise points marked, and cluster labels anchoring the drill-down (Figure 4.3). This is the same per-node embedding the faithfulness measures (Section 4.6) assess, so the scores describe exactly the view being navigated. Displayed alongside the projection, they flag a distorted view before the analyst drills in.

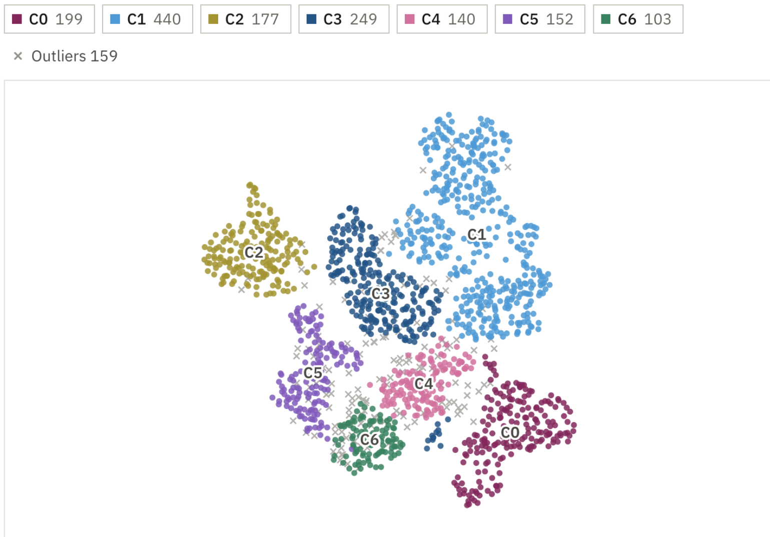


Figure 4.3.: The cluster-projection overview: the node's points in its own two-dimensional embedding, with each child cluster colored, noise points shown as gray crosses, and cluster labels anchoring drill-down. (Digits dataset [AK98].)

4.6. Configuration and Quality-Guided Selection

Different UMAP and HDBSCAN configurations produce different hierarchies. The hyperparameters are set once per run from manually chosen defaults, and apply to every expansion in that tree. The per-node faithfulness measures of Section 2.3 (trustworthiness, continuity, MRRE, stress, and CADI [GKM26; Jeo+23]) then indicate whether the regions the user navigates are trustworthy under that configuration.

Which configurations are worth choosing is an empirical question, so for the evaluation the choice is additionally treated as an optimization problem and searched with the Tree-structured Parzen Estimator (TPE) sampler of Optuna [Aki+19; Ber+11] under two objectives: *clustering quality*, measured by DBCV [Mou+14] in the original feature space, tuning the full pipeline including the embedding dimension; and *projection faithfulness*, measured by the mean of trustworthiness and continuity [LV09], tuning only the DR hyperparameters with the dimension fixed at two. The split follows from where each property lives: a good clustering is a property of the feature space, a faithful view a property of the embedding. Since every trial re-runs the full stack, the search is far too expensive for the interactive loop. It is an evaluation instrument, not a prototype feature, and was run once as reported in Section 6.5. A follow-up study

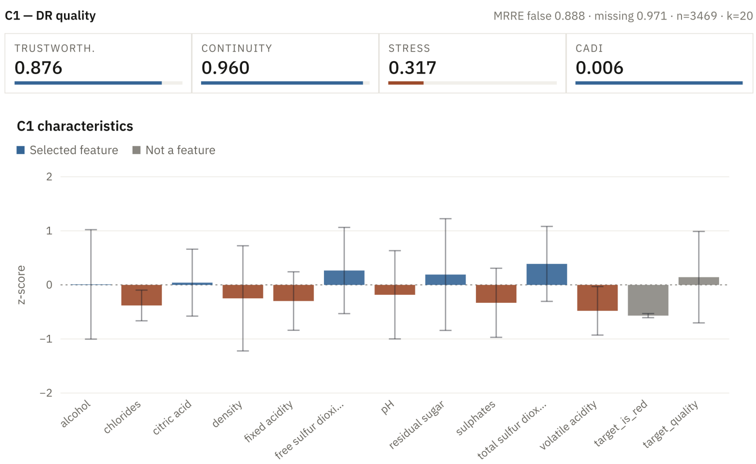


Figure 4.4.: The cluster-characteristics overview: the selected cluster's distributions and scores for each dimension. (Wine-quality dataset [Cor+09].)

tuning the deployed pipeline for per-dataset presets was halted before completion and is not reported here, so feeding tuned per-dataset configurations back into the defaults remains future work (Chapter 8).

4.7. Predicate Construction

A selected set of points is described by a conjunction of axis-parallel range constraints over the original features: the strict, DimBridge-style predicate [Mon+22; Mon+25] that serves as the baseline.

Let the analyst's selection be a set of pattern points P drawn from a background B (either the whole dataset or the current leaf, with scope discussed below). For each feature f_j the method forms a candidate interval clause $x_j \in [\,\text{sel\_min}_j,\ \text{sel\_max}_j\,]$ from the selected values, and a point satisfies the clause when its j-th feature falls in that interval. A predicate is a conjunction of such clauses. A point *satisfies the predicate* when it satisfies every clause. Write the selection membership as a binary label y (one for points in P, zero otherwise) and the predicate membership as a mask m. The quality of a predicate is then its agreement with the selection, measured by precision, recall, and their harmonic mean F1:

$$\text{precision} = \frac{|m \wedge y|}{|m|}, \qquad \text{recall} = \frac{|m \wedge y|}{|y|}, \qquad \text{F1} = \frac{2\,\text{precision} \cdot \text{recall}}{\text{precision} + \text{recall}}. \tag{4.1}$$

The strict predicate is built greedily by recursive predicate induction in the style of PIXAL's RPI [Mon+22]: starting from the empty conjunction, the method repeatedly adds the clause that most improves F1 and stops when none improves it further (improvement is tested against a 10^{-6} tolerance, and ties between equal-gain clauses are broken deterministically). PIXAL's original RPI scores expansions with a Bayes factor and maintains multiple candidate predicates. DimBridge re-scores the induction with F1 and likewise retains multiple overlapping candidate predicates [Mon+25]. The strict baseline here is thus a single-path, F1-scored approximation inspired by DimBridge's adaptation of PIXAL's RPI. The number of retained clauses is the predicate's *length*, a direct proxy for its complexity.

The *scope* of the comparison is itself an option. A global scope describes what makes the selection distinctive within the entire dataset, while a local scope describes what distinguishes it within the region already drilled into. The two answer different questions and can yield different predicates for the same selection.

4.8. Predicate Relaxation

Strict predicates take the full observed range of the selection per feature and can therefore be brittle: a few extreme selected points widen a clause, and small changes in the selection move the bounds. Relaxation trims each feature's range inward.

The amount of trimming is set by a single retained-center-mass (RCM) threshold $t \in (0, 1]$: a fraction $1 - t$ of the selection's probability mass is removed from each feature's clause, so $t = 1.0$ keeps every selected value (the strict predicate) and $t = 0.9$ discards the ten percent lying furthest in the tails. The novel element is *how* that trim is divided between the two tails. Rather than splitting it evenly, the method allocates it in proportion to each tail's *severity*, defined as the distance from the median to the extreme value on that side: the heavier, more outlier-stretched tail is trimmed more. The clause bounds are then read off as quantiles of the selected values,

$$\big[\ Q(\text{left\_trim})\,,\ \ Q(1 - \text{right\_trim})\ \big], \qquad \text{left\_trim} + \text{right\_trim} = 1 - t. \tag{4.2}$$

On finite selections the trim is quantized: the bounds are interpolated order statistics (`np.quantile` with linear interpolation), so on a tail with non-degenerate spread any $t < 1$ removes at least one selected point (a realized trim of at least $1/n$ on that tail, $2/n$ across both), while a tail whose median coincides with its extreme is not trimmed at all. Over the benchmark walk's leaves (n = 25–115), $t = 0.95$ removes roughly 5–10 % of the selected mass (median $\approx$ 6.7 %) rather than the nominal 5 %. The bounds themselves move continuously with t even where the set of excluded selected points does not. A relaxed bound is therefore generally an interpolated value rather than an observed one, whereas the strict bounds are exactly the selected minima and maxima. The intended effect is to retain the center mass of the subspace while discarding the

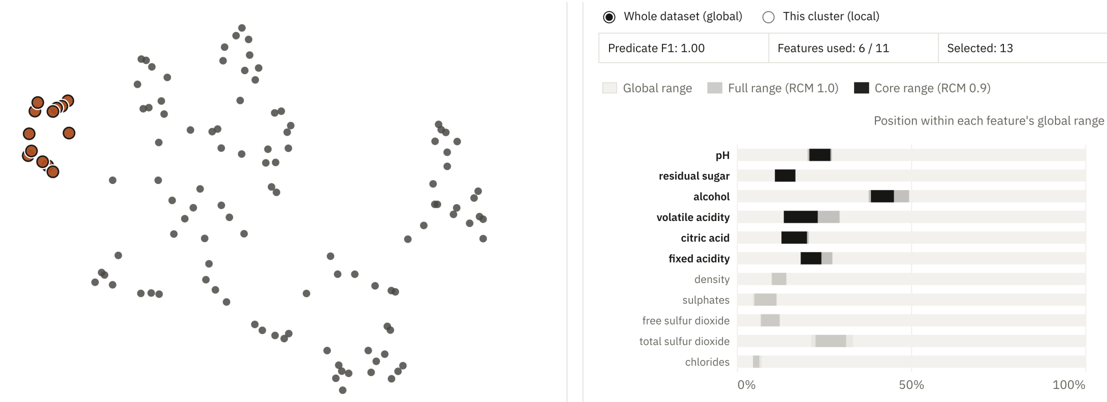


Figure 4.5.: The selection-and-predicate step: a lasso-selected point set (orange, left) is described per feature by its position within the feature's global range (right), the strict full range (RCM 1.0, gray) beside the relaxed core range (RCM 0.9, black); features the greedy induction did not select are grayed out. The header reports the predicate's F1, the features used, and the selection size, and the toggle switches between the global and the local scope. Wine-quality example, global scope.

width contributed by a few outlying points. The asymmetry ensures that a single extreme value on one side does not distort the opposite bound. Because the bounds then depend on the bulk of the selection rather than on its extremes, the description should change less when the selection changes, trading some specificity for robustness to the exact selection. This stability is the conceptual core of the relaxation method, and the property EQ4 puts to the test.

The two predicate modes are also complementary as a diagnostic. Because the strict predicate is the $t = 1$ case of the same construction, comparing a clause with its relaxed counterpart shows how much of the clause's width a few extreme points contributed, and on which side they lie (Figure 4.5): bounds that barely move are carried by the bulk of the selection, bounds that collapse by the tails. The reading holds per clause rather than for the predicate as a whole. In the greedy DimBridge variant [Mon+25], which decides *which* features enter the description, the candidate clauses are scored on the already trimmed ranges, so lowering t can change the selected feature set as well as its bounds. Section 6.8 reports the resulting drop in predicate length. Whether the comparison is a reliable sensitivity check in practice is not tested here. One caution on the term: in rule learning, relaxing a constraint usually means making it *less* restrictive, whereas the trim makes each interval narrower and the accepted region geometrically smaller. Relaxation here, following the project's terminology, relaxes the description's adherence to the exact selected extrema, not the predicate's acceptance. The price is paid in coverage and recall, and EQ4 measures it.

The implemented relaxation operator trims interval bounds. It does not drop whole clauses, although the project description also envisaged dropping narrow constraints. The greedy induction partially covers this by selecting which features enter the predicate at all. An explicit clause-dropping operator is left to future work (Chapter 8).

5. Implementation

5.1. System Overview

The HiLDE prototype is a client–server application: a Python analysis core with a separate node-scoring module, exposed through a thin FastAPI backend, and a React/D3 frontend. The two sides share one data contract: the backend serializes the analysis tree produced by the analysis core into a JSON schema, mirrored by the frontend's TypeScript interfaces. The core contains no UI code and can be driven headless, which makes the offline experiments of Section 5.5 possible. Figure 5.1 shows the two conceptual branches, the *hierarchical* decomposition of Chapter 4 and the *exploration* of whatever subset the analyst has reached, which share this data but run independently.

5.2. Analysis Layer

The analysis layer, called the *calc layer* in the code, is pure logic. Data is loaded as a pandas DataFrame with user-selected feature columns and an optional `StandardScaler` fit once at the root and reused. A reducer dispatcher provides PCA, t-SNE, and MDS via scikit-learn [Ped+11] and UMAP via `umap-learn`. A clustering dispatcher provides HDBSCAN (returning labels and GLOSH outlier scores), which drives the hierarchy. The dispatcher's K-Means and DBSCAN options are exercised by the tuning harness of Section 6.5. The interactive app pins HDBSCAN, the density-based choice of Section 4.5, and a Gaussian mixture model (GMM) is implemented but not exercised. The hierarchy itself is built by a recursive routine producing typed internal-node and leaf objects. Each node stores its original row indices, embedding (with PCA variance), and characteristics. Internal nodes additionally store outlier scores and their child layer, and quality scores are attached by a separate evaluation pass rather than by tree construction. The layer is navigated through three entry points: tree construction, dimensionality reduction, and clustering. Datasets enter through a registry of loaders, which wires the datasets of the evaluation (Chapter 6) and several further high-dimensional image and molecular benchmarks.

5.3. Predicate and Evaluation Layers

Predicates are produced by a single generator, which dispatches to the relaxed range method and the DimBridge-style greedy method. Per-feature bounds are computed as quantiles of the selected values, and the relaxation is the skew-aware tail trimming of Section 4.8 (total trim $1 - t$, split between tails by severity). The generator additionally returns per-clause and overall F1 and a per-point membership flag. Sections 4.7–4.8 give the conceptual detail. Node quality is scored by the evaluation module, which computes stress, trustworthiness/continuity, MRRE and CADI with ZADU [Jeo+23]. For interactive use, the prototype recomputes trustworthiness/continuity, MRRE, and stress from a chunked variant of the same formulas that keeps large nodes within memory, verified equivalent by checks shipped with the repository, with ZADU still supplying CADI at internal nodes. Both modules guard against degenerate inputs, returning empty predicates for empty selections and `None` scores for nodes too small to embed or score reliably. Predicate fidelity against ground truth and stability under perturbation are covered by the offline harness of Section 5.5 and evaluated in Section 6.8.

5.4. User Interface and Workflow

The interface runs the workflow end to end: global configuration → cluster-projection overview and drill-down (clicking a cluster descends one level) → leaf exploration → lasso/box selection → predicate and range analysis → CSV export. The React/D3 frontend renders these views. The FastAPI backend serves the analysis tree and the predicate results from the analysis core through the serialized data contract of the system overview.

For experiments that must be repeatable, the same analysis layer can be driven without the interface. The scripted "simulated user" of the stability harness (Section 5.5) builds the tree headless, takes leaf clusters (or, on synthetic data, planted clusters) as the "selected" set in place of a manual lasso, and perturbs these selections programmatically. Removing the human selection step is what makes the predicate-stability experiment (EQ4) reproducible.

5.5. Offline Experiment Harness

Experiments that do not run through the interactive app live in a separate research package: one harness per experiment (the tuning-effectiveness study, the hierarchical-versus-flat comparison, the nested-subspace experiment, and the predicate-stability experiment). All four drive the unchanged calc layer and write their records, summaries and plots to a time-stamped run directory in the repository: the EQ1 and EQ2 harnesses run the shared reducer/cluster dispatch over the dataset registry, while the EQ3 and EQ4 generators build their planted data and call the pipeline pieces they test directly. A fifth harness drives the benchmark walk of Section 6.9. A sixth harness implements a follow-up preset-tuning study that was halted before completion and is not reported in Chapter 6. Seeding policy and the shared statistical protocol are stated once, with the experimental setup (Section 6.3).

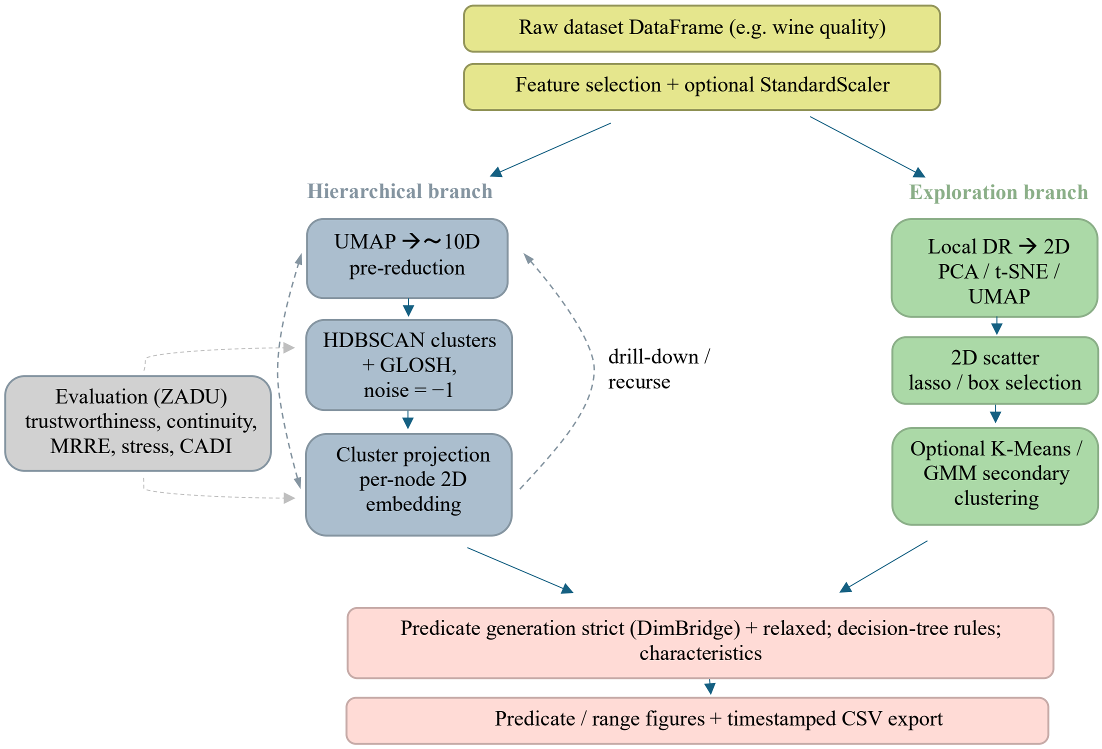


Figure 5.1.: The HiLDE pipeline. After feature selection and optional scaling, the data feeds two branches that share data but run independently: the *hierarchical* branch (UMAP → HDBSCAN → per-node cluster projection, recursed via drill-down) and the *exploration* branch (a user-chosen 2D projection with lasso selection). Three details of the drawing are idealized: the secondary-clustering dispatch belongs to the analysis layer and is exercised by the offline harness on the global embedding, not by the shipped UI, and, despite the drawn arrow, does not feed the predicate step; the predicate step produces range predicates only (the decision-tree annotation is an option that did not ship); and the UMAP pre-reduction is skipped under the app defaults, which cluster in the full feature space (where the evaluation's pre-reducing variant is used, it reduces to two components rather than the ten shown). Both branches feed the predicate and characterization step. Projection-faithfulness scores are attached to every node.

6. Evaluation

This chapter reports the empirical evaluation. Section 6.1 states four evaluation questions, one per experiment. Three sections then fix the shared ground: the datasets, the experimental setup, and the metrics. Sections 6.5 to 6.8 report the four experiments, each stating its setup and results before interpreting them and closing with a verdict on its question (EQ2's verdict is issued jointly with EQ3's). Section 6.9 checks the shipped defaults against predictions derived from those experiments on real benchmarks, and Section 6.10 adds a case study and expert feedback. The verdicts are collected once more in Section 8.2.

6.1. Evaluation Questions

The evaluation is organized around four questions, one per experiment. Each is specific to the approach of Chapter 4: it names the mechanism under test, the baseline, and the criterion that decides it. EQ1 tests the configuration of the decomposition pipeline, EQ2 and EQ3 the hierarchical decomposition itself (the first aim of Section 4.1), and EQ4 the predicate-relaxation layer (the second aim). The measures throughout are those of the third aim.

EQ1 (Configuration, Section 6.5). Does automated hyperparameter search (TPE under the two objectives of Section 4.6) improve the pipeline's density-clustering validity (DBCV) and the two-dimensional embedding's faithfulness (trustworthiness and continuity) over the prototype's default configuration, and does optimizing these internal objectives also recover externally labeled structure? The protocol itself is on trial here: if a measure ranks configurations in a way unrelated or opposed to the known ground truth, and the divergence cannot be attributed to a specific, checkable mechanism, that counts against the protocol, not as information.

EQ2 (Local views, Section 6.6). For a region reached through the recursive decomposition (Section 4.3), is the region's own two-dimensional re-embedding a more faithful view of its structure than the same points read off a single global projection (per DR method, and under a method-neutral control for how regions are chosen), and does the resulting leaf partition recover ground-truth classes better than a single flat HDBSCAN? The two halves are stated as hypotheses H1a and H1b with the experiment.

EQ3 (Nested structure, Section 6.7). When fine sub-cluster structure is separable only *after* conditioning on a coarse group (planted as nested, per-group-rotated subspace clusters), does the recursion recover it where a single flat clustering merges it by construction (H1c), and does the advantage disappear on a non-nested control where the predicted mechanism is absent (H1d)?
EQ4 (Relaxation, Section 6.8). Does trimming each clause inward by the retained-center-mass threshold (Section 4.8) make the description of a selection more stable under small perturbations of that selection, at a specificity cost gradual enough that a favorable operating point exists, and does the severity-proportional tail split, the method's stated novel element, outperform a naive symmetric split? The pre-specified hypothesis H2 and the operating-point criterion are stated with the experiment.

6.2. Datasets

Evaluation uses two families of data. The first is *synthetic* data with *planted* subspace clusters, whose known generating rule gives a ground truth against which the recovered structure (EQ3) and the predicates (EQ4) can be scored with precision and recall. Each synthetic dataset is specified by its point count, dimensionality, cluster count, the relevant dimensions and their ranges per cluster, the number of pure-noise dimensions, and the degree of cluster overlap. The second family is *real benchmark* datasets, used to show behavior where no ground-truth subspace exists. Here the evaluation is descriptive and faithfulness-based.

The datasets used in the evaluation are listed in Table 6.1, a subset of the prototype's ten-loader registry: two synthetic generators, concentric rings (non-convex density clusters with ground-truth labels) and a Swiss roll (a continuous manifold without discrete classes), alongside four real tabular and image benchmarks. The tuning experiment (Section 6.5) runs on the four bold entries, chosen to contrast low- and high-dimensional, convex and non-convex, labeled and density-only data.

The selection varies along the axes the method should be sensitive to. Dimensionality ranges from four features (iris) to 64 (digits). The concentric rings contribute non-convex density structure with known labels, the case centroid-based methods mishandle and density-based clustering should not. The Swiss roll separates "faithful embedding" from "recovered clusters". The real tabular sets carry interpretable features so that example predicates can be read and sanity-checked. To keep the neighbor-based reductions (t-SNE, UMAP) and the $O(n^2)$ faithfulness measures tractable, each dataset is subsampled with a fixed seed: to 500 rows in the tuning experiment and 1,000 rows in the later experiment harnesses.

Beyond the registry's synthetic generators, two experiment-specific generators plant

Table 6.1.: Datasets used in the evaluation, a subset of the prototype registry, under their registry names, whose (Low) suffix marks the registry's lower-dimensional group. The registry's remaining, higher-dimensional image and molecular sets are not covered by any experiment. Bold entries are used by the tuning experiment. Concentric rings and Swiss roll are synthetic, and the rest are real.

Dataset	Type	# Features	Notes
Wine quality (Low)	real tabular	11	interpretable, red/white ground truth
Breast cancer (Low)	real tabular	30	classic, 2 classes
Digits (Low)	real image	64	10 classes, manifold structure
Concentric rings (Low)	synthetic	2	non-convex density, ground-truth labels
Iris (Low)	real tabular	4	small, very interpretable
Swiss roll (Low)	synthetic	3	continuous manifold, no classes

known structure. The planted-subspace experiment (Section 6.7) plants nested, per-group-*rotated* subspace clusters. The predicate-stability experiment (Section 6.8) plants *axis-parallel* subspace clusters, from which predicate *recovery* (precision/recall of the relevant-dimension set) can be scored. The two designs must differ: the rotation that defeats the flat baseline in the EQ3 generator is structure that axis-parallel range predicates cannot express, so the EQ4 generator is axis-aligned.

6.3. Experimental Setup

Unless stated otherwise, all experiments share the following setup. Features are z-score standardized, with means and variances estimated once on the full dataset before any reduction. The UMAP pre-reduction is swept over its neighborhood size (`n_neighbors`), minimum distance (`min_dist`), and component count (`n_components`). HDBSCAN is governed by `min_cluster_size` and `min_samples`. Predicate relaxation is evaluated across the thresholds $t \in \{1.0, 0.95, 0.9, 0.8\}$, with $t = 1.0$ recovering the strict baseline. Random seeds are fixed (seed 42) for subsampling and the Optuna studies. At the time the experiments ran, the reducers passed no internal seeds (t-SNE, UMAP, and MDS all unseeded). PCA is deterministic. Under the pinned scikit-learn version (1.9.0), repeated unseeded t-SNE runs reproduced identical embeddings at the node sizes and feature counts of this evaluation (verified post hoc), an empirical property of these runs rather than a documented guarantee, so the PCA and t-SNE arms are treated as single effective draws. UMAP and MDS vary across runs. The stability and benchmark experiments (Sections 6.8 and 6.9) report means over repeated builds. The tuning study of Section 6.5 is a single study per cell. The released prototype has since seeded all three reducers. Section 6.10 discloses the change. Selections for the predicate experiments come from the scripted simulated user of Section 5.5.

Reproducibility. The prototype, the experiment harnesses, and all run directories are part of the project repository (tag `final-thesis`, Appendix A). All experiments ran in a single pinned Python environment. The exact version of every package is recorded in the repository's lock file. Each experiment writes its raw records and summaries as CSV files to a time-stamped run directory in the repository, from which every number reported in this chapter is taken. Where a run directory carries a corrected re-derivation, the corrected value is reported. Where a hypothesis, reporting rule, or decision criterion is described below as pre-specified, it was fixed before execution and implemented in the experiment harness rather than selected after inspecting the results.

Statistical protocol. Unless a section states otherwise, the analyses share one protocol. The experimental unit is the region or the selection, never the perturbation replicate: replicates are aggregated to one value per unit before any test (Section 6.8). Paired conditions on the same units are compared with the Wilcoxon signed-rank test [Wil45] at $\alpha = 0.05$. Where a family of relaxation thresholds is tested jointly, p-values are Holm-corrected [Hol79] (Section 6.8). Effects are reported as median paired differences with win rates, and rank-biserial correlations where stated. Confidence bands in figures are 95% intervals over builds. Independent replicates per experiment: five unseeded rebuilds in EQ2 (identical draws in the deterministic PCA and t-SNE arms) and in the benchmark walk, twenty generator seeds in EQ3 and in EQ4's synthetic arms.

6.4. Metrics and Their Justification

Each metric is chosen because it answers a specific evaluation question. The set is grouped by what it measures.

Projection faithfulness (EQ1–EQ3) is measured per node with the ZADU measures [Jeo+23]: trustworthiness and continuity (T&C), MRRE, and stress. They quantify whether the neighborhoods and distances the user sees reflect the original space [LV09], the core claim behind EQ2. Trustworthiness/continuity and MRRE are local (neighborhood-preservation) measures and stress is global (distance-preservation). CADI, the cluster-level member of the same family [GKM26], completes the local/global/cluster axes in the dashboard's per-node scores (Section 5.3) but is not reported by the experiments of this chapter. Trustworthiness, continuity, and the inverted MRRE are higher-better, while stress and CADI are lower-better. CADI is non-negative with optimum 0, and ZADU reports it normalized to $[0, 1]$. DBCV [Mou+14] is a density-aware internal cluster-validity index that, unlike compactness-based indices such as the silhouette score, rewards well-separated clusters of arbitrary shape. It serves as the clustering-quality objective of the tuning experiment (Section 6.5).

Predicate quality (EQ4) is measured by precision, recall, and F1 of predicate membership against the target set, plus *coverage* (the fraction of the background, the whole dataset or the current leaf according to the predicate's scope, that the predicate admits) and *predicate length* (number of clauses). The first three test whether the description still separates the intended points. Coverage and length are proxies for generality and interpretability, and together with precision and recall they locate a description on the specificity-versus-robustness trade-off that relaxation is meant to move.

Stability (EQ4, the central claim) is measured by the variability of the description under perturbation: the Jaccard overlap, or adjusted Rand index (ARI), of the predicate-selected sets, and the standard deviation of predicate F1, under (a) small changes to the user selection and (b) independent unseeded rebuilds of the hierarchy, matched across builds by member overlap (Section 6.8). This directly tests the stability clause of EQ4. Faithfulness and predicate-quality alone do not capture it.

Recovery (synthetic only, EQ3/EQ4) is the precision and recall of the recovered relevant-dimension set and of the planted membership.

Label agreement at two granularities (EQ3, and descriptively in the benchmark walk) is reported as homogeneity, completeness, and their harmonic mean V-measure [RH07], with a fragmentation ratio (clusters per ground-truth group) separating over-splitting from mixing. The benchmark walk of Section 6.9 additionally summarizes leaves by purity and label enrichment against the withheld class column.

For reference, the Jaccard index of two sets A and B, $|A \cap B|/|A \cup B| \in [0, 1]$ (higher better), is used to compare two predicate-admitted sets. The ARI compares two partitions corrected for chance [HA85], taking the value 1 for identical clusterings, around 0 for random agreement, and possibly negative values for worse-than-random agreement. The MRRE components are rank-*error* measures [LV09], reported here in their inverted form ($1-$error, as returned by ZADU [Jeo+23]).

6.5. Hyperparameter-Tuning Effectiveness Experiment (EQ1)

This experiment answers EQ1: it measures *how effective* automated hyperparameter tuning is for the DR$\rightarrow$clustering pipeline. It is built around two objective *tracks* and four *effectiveness axes*, evaluated over a grid of datasets, DR methods, and clustering methods.

Tracks. Track A (*clustering quality*) optimizes DBCV [Mou+14] in the original space, tuning the full pipeline (DR hyperparameters, embedding dimension, clustering hyperparameters) across dataset × DR × clustering. DBCV is distance-based and so affected by distance concentration in high dimensions, so its absolute values are read conservatively below: comparisons rest on differences between configurations, and a score's sign indicates only whether a partition is density-separated at all. Track B (*DR faithfulness*) optimizes the mean of trustworthiness and continuity [Jeo+23; LV09] of a two-dimensional embedding, tuning the DR hyperparameters alone, so the clustering axis collapses.

Effectiveness axes. Every grid cell is quantified along four axes: (1) *default-vs-tuned gain*, the best tuned objective minus the score at the project defaults; (2) *TPE vs. random search*, an Optuna TPE study against a random-search study of equal budget [Aki+19; Ber+11], isolating *intelligent* search from luck; (3) *convergence speed*, the best-so-far curve over trials; and (4) *external validation*, agreement of the tuned clusters with ground-truth labels (ARI, normalized mutual information (NMI)), testing whether optimizing an internal index yields *real* structure rather than a metric artifact.

Setup of the completed run. The grid ran over the four datasets of Table 6.1 (run `20260628_153633`) with DR ∈ {UMAP, t-SNE, PCA, MDS} and, for Track A, clustering ∈ {HDBSCAN, DBSCAN, K-Means}, giving 64 grid cells. In 60 of them one TPE and one random-search study of 100 trials each were run (seed 42) on a 500-row seeded subsample, the four Track-B PCA cells having no tunable parameters and being evaluated at their deterministic configuration only. Search ranges: UMAP $n_{\text{neighbors}} \in [5, 50]$, min_dist $\in [0, 0.5]$, components $\in [2, \min(d, 15)]$; t-SNE perplexity $\in [5, 50]$, learning rate $\in [10, 1000]$; PCA components $\in [2, d]$; MDS components $\in [2, \min(d, 10)]$; HDBSCAN min_cluster_size $\in [2, 50]$, min_samples $\in [1, 25]$; DBSCAN $\varepsilon \in [0.1, 5]$; K-Means $k \in [2, 15]$. The shipped defaults these gains are measured against are the configuration of Section 6.6's trees (UMAP 15/0.1, t-SNE perplexity 30, HDBSCAN 25/5, DBSCAN $\varepsilon = 0.5$, K-Means $k = 5$). Clusterings with fewer than two clusters or more than 50% noise were scored as the worst objective value. Because the studies ran on the 500-row subsample, the tuned `min_cluster_size` and `min_samples` are absolute point counts and do not transfer across dataset sizes, a limitation identified after the run, in the design of a follow-up preset-tuning study that was not completed (Section 5.5).

Results. First, *tuning matters greatly for clustering quality but hardly at all for projection faithfulness.* On Track A the project defaults are almost universally poor (negative DBCV) and tuning moves many configurations into positive territory (Table 6.2, per-cell gains

Table 6.2.: Track A tuned results (TPE), averaged over datasets and DR methods, by clustering method, together with the mean best objective per track for TPE vs. random search. DBCV $\in [-1, 1]$ (higher better), ARI higher better, T&C $\in [0, 1]$.

Track A clustering	mean best DBCV	mean gain	mean ARI
HDBSCAN	0.17	0.30	0.34
DBSCAN	0.16	0.61	0.23
K-Means	−0.21	0.24	0.32
Mean best objective: TPE vs. random			
Track A (DBCV)		TPE 0.040	random 0.001
Track B (T&C)		TPE 0.950	random 0.949

in Figure 6.1). These gains are in-sample by construction (the objective is evaluated on the data it was optimized on). On Track B the defaults already sit near the ceiling (trustworthiness/continuity $\approx$ 0.88–1.00) and tuning adds only $+0.006$ on average: those embeddings are already about as faithful as the data allows.

Second, *intelligent search beats random search only when the landscape is hard.* TPE reaches a higher mean best objective than random search on Track A (0.040 vs. 0.001) but is essentially tied on Track B (Table 6.2). Both samplers make most of their progress in the first $\sim$20 trials, with TPE pulling ahead later on the harder Track A objective (Figure 6.2a).

Third, *the clustering algorithm dominates the DBCV outcome.* Averaged over datasets and DR methods, the density-based methods reach a positive mean best DBCV while K-Means stays negative (Table 6.2), as expected from a density-aware index that penalizes convex partitions. The largest DBCV *gains* (e.g. t-SNE+DBSCAN, or PCA/MDS+DBSCAN on the rings) are partly an artifact of degenerate defaults scored at -1.0, so gains should be read alongside the absolute best.

Fourth, *optimizing an internal index does not reliably recover the labeled classes.* Across Track A the correlation between a cell's DBCV gain and the tuned clustering's ARI is weak ($r \approx 0.17$, Figure 6.2c), and K-Means attains an average ARI comparable to HDBSCAN's despite its strongly negative DBCV (Table 6.2). Only tuned configurations were scored against the labels, and no default-configuration ARI was recorded, so this is a correlational reading rather than a measured before–after gain in label recovery. This is a concrete instance of "classes are not clusters" [Jeo+24]: DBCV rewards density-separated structure, which need not coincide with the ground-truth labels.

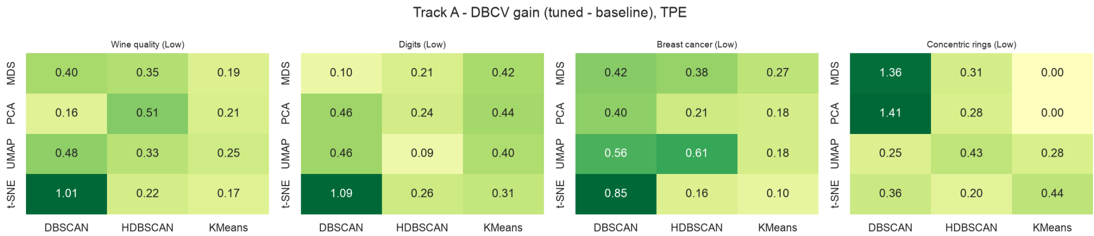


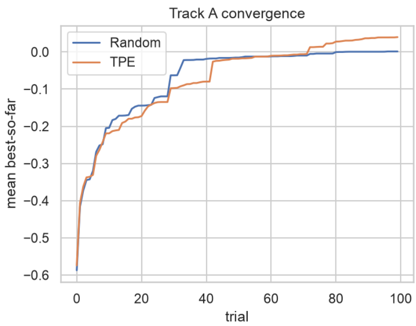
Figure 6.1.: Track A DBCV gain (tuned − default, TPE) per DR × clustering cell, faceted by dataset. Gains are positive almost everywhere. The darkest cells (t-SNE/PCA/MDS with DBSCAN) are inflated by degenerate defaults scored at −1.0.

Verdict (EQ1). A partial pass under the rule stated with the question. The faithfulness half passes: view faithfulness sits near its ceiling untuned and the measures discriminate configurations. The validity half passes only as an in-sample result: tuning lifts DBCV substantially, partly off degenerate defaults scored at the floor, and the DBCV gains tuning finds are nearly uncorrelated with agreement to the labeled classes ($r \approx 0.17$, $n = 48$, $p = 0.26$), a divergence attributed to the general “classes are not clusters” phenomenon rather than to a mechanism this experiment checks.

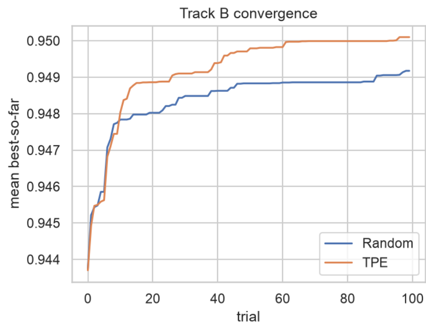
6.6. Hierarchical versus Flat Embeddings (EQ2)

This experiment answers EQ2. The question splits into two claims that need different baselines: *H1a*, that re-embedding a region on its own preserves that region’s internal structure better than reading the same points off a single global projection; and *H1b*, that the recursive decomposition recovers a known ground-truth partition better than a single, non-recursive (flat) clustering. The two are kept apart because the results point in opposite directions.

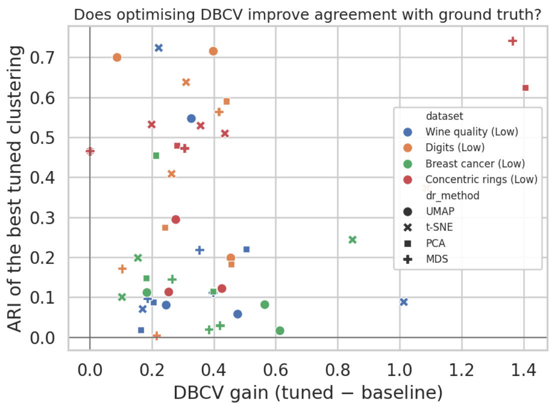
Setup. The grid covered five datasets (concentric rings, wine quality, breast cancer, digits, and the Swiss roll, the last used for H1a only), four DR methods {UMAP, t-SNE, PCA, MDS}, and five unseeded rebuilds, with the hierarchy depth fixed at two, giving 100 grid cells (run `20260628_184827`). Trees are built under the shipped defaults (pre-reduction to two UMAP components, HDBSCAN at minimum cluster size 25 and minimum samples 5), and the neighborhood size forced equal below is $k = \min(20, \lfloor (n-1)/2 \rfloor)$ for a region of n points. For H1a the comparison is *paired* per region: the same point set is scored once on the leaf’s own local embedding (*hierarchical*) and once on a single global embedding of the whole dataset sliced to those rows



(a) Track A convergence.



(b) Track B convergence.



(c) Internal (DBCV) vs. external (ARI).

Figure 6.2.: (a, b) Mean best-so-far objective per trial: on the harder Track A objective TPE overtakes random search late, while on Track B both plateau almost immediately. (c) DBCV gain barely predicts the tuned clustering's ARI ($r \approx 0.17$). The vertical axis is the tuned run's ARI itself, not an ARI change. Track A records no baseline ARI.

(*flat*), both using the identical scoring path so that the neighborhood size k is forced equal. To separate a genuine effect from leaves being self-selected compact regions, H1a is run under two region definitions: the hierarchy's own leaves (*hier_leaf*) and, as a method-neutral control, the ground-truth classes (*gt_class*). For H1b the hierarchical leaf partition is compared against one non-recursive HDBSCAN run on the same preprocessed space, both scored against the ground-truth labels by ARI and NMI. Noise is treated symmetrically: ARI and NMI are computed over the rows labeled non-noise by *both* partitions, and each partition's own noise fraction is reported alongside (Table 6.4), so recovery is compared on the subset both methods actually label.

H1a: faithfulness of local views. The result is strongly DR-method-dependent (Figure 6.3, Table 6.3). For the projection methods that target global structure, the local re-embedding is significantly more faithful: aggregated across datasets, PCA and MDS gain median per-region trustworthiness on hierarchy leaves and win in roughly four regions in five ($p \ll 10^{-3}$, Wilcoxon; Table 6.3). The concentric-rings cell is degenerate for the linear methods: the data are two-dimensional, so a two-dimensional linear re-embedding reproduces the global view exactly and all 90 leaf deltas are zero, entering the mean win rates as ties counted against the local view. Excluding it, PCA and MDS gain trustworthiness in 99% of regions. These gains are not bought at the expense of global distance: for PCA and MDS the local embeddings also reduce *stress* in 82–98% of regions, so they are more faithful on both the local and the global axis at once. The stress comparison is read for these two methods only: the implemented stress is sensitive to a uniform rescaling of the embedding, and t-SNE and UMAP coordinates carry no meaningful scale, so their stress values conflate scale with distortion and are not interpreted. For UMAP the trustworthiness gain is small, so the local view is roughly a wash. For t-SNE the comparison *reverses* on the neighborhood measures themselves: the local re-embedding is markedly worse, winning only $\sim 10\%$ of regions.

Two mechanisms explain this pattern. A global PCA or MDS axis is fixed by whole-dataset variance and can be almost orthogonal to a compact region's own variation, whereas a local re-fit adapts to it. The t-SNE reversal, in turn, is best explained as a small-region artifact combined with a ceiling effect. A global t-SNE already optimizes local neighborhoods, so a region read off it is *already* locally faithful and there is little to gain. Re-running t-SNE on a small leaf with an unadapted perplexity is known to produce unstable, globally distorted embeddings [WVJ16]. The control supports this reading: under the method-neutral *gt_class* regions, which are larger and not chosen by the hierarchy, the t-SNE deficit essentially vanishes while the PCA and MDS advantages fall to roughly a third. Aggregating the five builds to one value per class (the region-level unit declared above), the advantage remains significant on digits ($p = 0.006$ PCA, $p = 0.002$ MDS; ten classes), while wine quality and breast cancer contribute only two class regions each, so no two-sided significance is attainable at that unit and only sign and magnitude are read (median Δ $+0.021$ to $+0.044$, positive for both methods; Table 6.3). On the concentric rings the control effect is nil (PCA deltas exactly zero, MDS $p = 0.95$). The reduction also quantifies the conditioning caveat: about two thirds of the leaf-region advantage is attributable to leaves being self-selected compact regions, but a real, method-neutral advantage survives for the global DR methods.

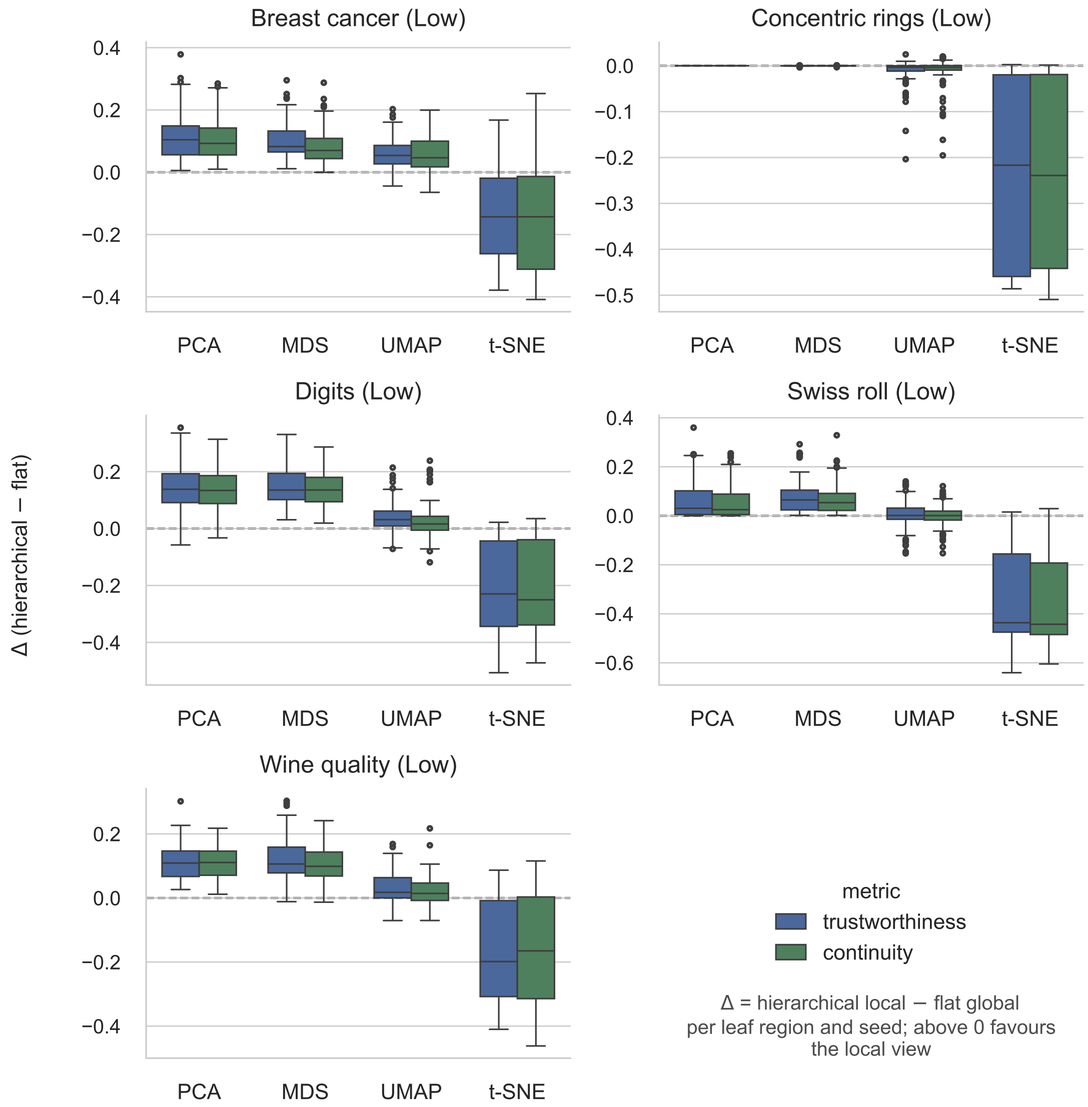


Figure 6.3.: H1a paired deltas (hierarchical local − flat global) per DR method, one panel per dataset, for both primary metrics (trustworthiness, continuity) on hierarchy-defined regions. Each box collects one delta per leaf region and rebuild; a box above the dashed zero line means the local view is more faithful. PCA and MDS show consistent positive gains on every dataset except the two-dimensional rings, where the linear methods are degenerate. UMAP is near zero, and t-SNE is strongly negative (cf. the control in Table 6.3). Note the per-panel y scales.

Table 6.3.: H1a per-region trustworthiness delta (hierarchical − flat), aggregated across datasets as the median of per-(dataset) median deltas, with the mean win rate. Positive favors the local (hierarchical) view. *hier_leaf* uses hierarchy-defined regions, while *gt_class* is the method-neutral control. The pattern for continuity is the same. The control's inferential unit is the class (see text). Its medians and win rates are descriptive over builds.

	hier_leaf		gt_class (control)	
DR method	median Δ	win rate	median Δ	win rate
PCA	+0.104	0.79	+0.032	0.73
MDS	+0.082	0.81	+0.029	0.83
UMAP	+0.018	0.65	+0.006	0.69
t-SNE	−0.217	0.10	−0.002	0.43

Table 6.4.: H1b ground-truth recovery (mean over DR methods and rebuilds): ARI and NMI of the hierarchical leaf partition versus a single flat HDBSCAN run, and the noise fraction of each. The flat clustering matches the labels better on every dataset.

Dataset	ARI_h	ARI_f	NMI_h	NMI_f	noise_h	noise_f
Concentric rings	0.33	1.00	0.58	1.00	0.20	0.00
Wine quality	0.25	0.91	0.40	0.88	0.13	0.01
Breast cancer	0.41	0.77	0.46	0.68	0.23	0.02
Digits	0.59	0.78	0.80	0.87	0.16	0.01

H1b: ground-truth class recovery. Here the result is a clean negative (Table 6.4). A single flat HDBSCAN matches the labels far better than the hierarchical leaf partition on every dataset, and flat wins in 99% of cells. The gap is widest where the hierarchy was expected to help: on the concentric rings a single run recovers the three rings perfectly while the hierarchy splits each ring into arcs. The hierarchical partition is also less stable across rebuilds (on wine, ARI ranges 0.09–0.85, standard deviation 0.25) and labels more points as noise (Table 6.4).

The cause is over-segmentation: recursion keeps subdividing a class across several leaves, and ARI penalizes that splitting, consistent with the documented tendency of UMAP-style embeddings to create false tears that yield a finer clustering than the data contain [MHM18b].

Taken together, the two halves show that the hierarchy's contribution is *representational* rather than *partitional*. It offers more faithful local views when the projection method is global or linear, but it does not produce a better flat partition of the data.

6.7. Planted-Subspace Recovery (EQ3)

The H1b negative of the previous section is incomplete, because benchmark class labels cannot express the one structure the recursive decomposition is built for: structure that lives in a *subspace* and is globally hidden but conditionally visible. Real classes are coarse partitions with no planted subspace nesting, so a flat clustering that already resolves them leaves the hierarchy nothing to add. This experiment supplies that structure under controlled conditions and asks: when fine structure is separable only *after* conditioning on a coarse group, and only along a per-group feature direction, does recursion recover it where a single flat clustering cannot?

Setup. A synthetic generator plants *nested* subspace structure (run `20260628_195214`): $G = 3$ coarse groups, each split into $K = 4$ fine sub-clusters (12 planted clusters, 60 points each), in $D = 16$ dimensions partitioned into a level-1 block ($d_A = 3$), a level-2 block ($d_B = 3$), and $d_{\text{noise}} = 10$ isotropic-noise dimensions. The coarse groups are separated along block A at a scale ρ. Within each group the fine sub-clusters are separated along block B but *rotated per group*, so the fine signal cancels when pooled across groups and is visible only locally. The headline knob is ρ, the coarse-vs-fine scale ratio, swept over $\{1, 2, 4, 8, 16, 32\}$. The unit of analysis is the planted fine label. Four conditions share an identical HDBSCAN (`min_cluster_size` = 15), so only recursion and subspace scope vary: *hierarchical*, the tree's leaf partition; *flat_full*, one HDBSCAN on the full standardized space (the same baseline as H1b); *flat_oracle_B*, one HDBSCAN on block B only, testing whether knowing the fine subspace globally suffices; and *oracle_conditional*, HDBSCAN on block B within each true group, an oracle-informed reference for the recursive idea. The grid covered the six ρ values, a nested and a non-nested regime, and 20 seeds (240 cells), with UMAP as the clustering DR method and hierarchy depth fixed at two. The headline metric is *within-group ARI* (ARI of the predicted against the planted fine labels, computed inside each true group over the condition's non-noise rows and averaged), which isolates the level-2 recovery from the coarse split.

Baseline behavior. Both *hierarchical* and *flat_full* recover the coarse groups perfectly (coarse ARI = 1.00 at every ρ), so any difference is at the fine level. *flat_oracle_B* fails to recover the coarse split (coarse ARI 0.37 vs. 1.00), expected since block B carries none by construction, but holds within-group ARI 0.647 (88 % of the conditional oracle) at every ρ, overtaking the hierarchy from $\rho = 4$.

Table 6.5.: Within-group ARI of fine-label recovery in the *nested* regime: hierarchical versus flat_full, per scale separation ρ (mean over 20 seeds). Δ is the paired median (hierarchical − flat). Win rate and Wilcoxon p are over seeds, and oracle-relative is the hierarchical within-group ARI as a fraction of the conditional oracle's (0.733 at all ρ).

ρ	hier	flat	median Δ	win rate	Wilcoxon p	oracle-rel.
1	0.700	0.586	+0.113	0.65	0.015	0.95
2	0.654	0.523	+0.109	0.60	0.033	0.89
4	0.578	0.442	+0.104	0.75	0.011	0.79
8	0.486	0.336	+0.127	0.80	0.005	0.66
16	0.439	0.289	+0.170	0.70	0.004	0.60
32	0.413	0.284	+0.139	0.75	0.006	0.56

H1c: within-group recovery. The hierarchy beats the flat baseline on within-group recovery at *every* scale separation (Table 6.5, Figure 6.4), with median deltas of +0.10 to +0.17 and Wilcoxon $p < 0.05$ throughout. The over-segmentation signature from H1b is still present but now works *for* the hierarchy rather than against it: the flat clustering under-segments (fragmentation ratio < 1, completeness ≈ 0.99 but homogeneity only 0.62–0.78), i.e. it lumps the fine sub-clusters of a group into one cluster, while the hierarchical leaves are balanced ($h \approx c \approx$ 0.75–0.87, fragmentation ≈ 1.2). They find the fine structure and split it only mildly (Figure 6.5). The global V-measure of the two methods *converges* as ρ grows (0.743 versus 0.742 at $\rho = 32$): the global score is dominated by the coarse split that both methods recover, and only the within-group metric keeps the level-2 difference visible. Conditional scoring thus controls the confound that made H1b's ARI misleading.

H1d: non-nested control. Because the generator was built around the mechanism the method has (sample-conditioned multi-scale separation, not feature selection), a hierarchy win on nested data is partly by construction. The non-nested control guards against reading an artifact as evidence. With the same 12 clusters placed at a single level (no coarse$\rightarrow$fine nesting), the hierarchy loses its rationale: the median paired hierarchical−flat within-group gap is +0.127 in the nested regime (mean over the six ρ) but −0.063 in the non-nested control, where the hierarchy is significantly *worse* than flat (condition means 0.88 vs 0.92; Wilcoxon $p = 0.019$, rank-biserial −0.5; the control is ρ-invariant by construction, so it is a single condition rather than a sweep), the over-segmentation cost with none of the conditioning benefit. The advantage therefore appears only where the predicted mechanism applies.

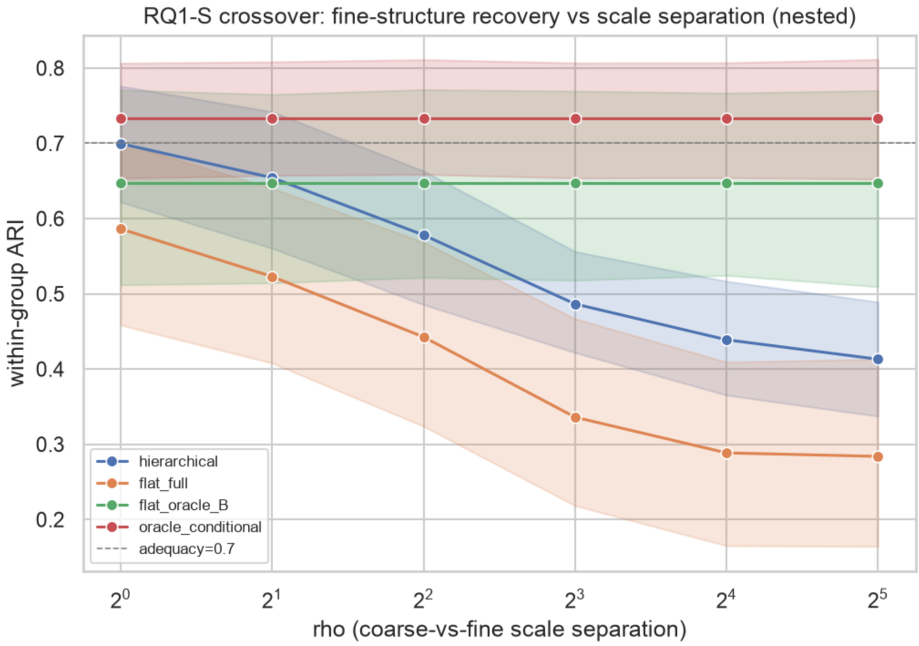


Figure 6.4.: Within-group ARI of fine-structure recovery against scale separation ρ (nested regime, mean $\pm$ 95% band over 20 seeds). The hierarchical partition (blue) stays above the flat baseline (orange) at every ρ. Neither reaches the conditional oracle (red), and the global-subspace baseline (green) is constant across ρ, overtaking the hierarchy from $\rho = 4$. The horizontal line is the design's 0.7 adequacy threshold.

Limits: no crossover, and a widening gap to the oracle. The design predicted a *crossover*: flat adequate at low ρ, failing at high ρ. Instead, flat is below the 0.7 adequacy threshold already at $\rho = 1$, so the estimated crossover ρ^* degenerates to 1.0 and the result is a *monotone divergence*: both methods degrade as the coarse scale grows, but flat degrades faster and the gap widens. The second limit: the hierarchy's share of the fine structure the conditional oracle recovers falls from 95% at $\rho = 1$ to 56% at $\rho = 32$ (Table 6.5). The oracle clusters block B alone, whereas the hierarchy clusters the full within-group space (block B plus residual block A and the ten noise dimensions). As the global standardization is set increasingly by the coarse scale, that within-node space becomes a noisier place to resolve the fine clusters. The method recovers structure a flat pass cannot, but it leaves a growing margin on the table because it cannot *select* the relevant subspace, a quantified motivation for a feature-weighting node splitter (Chapter 7). A third limit concerns the scoring rule: within-group ARI is computed

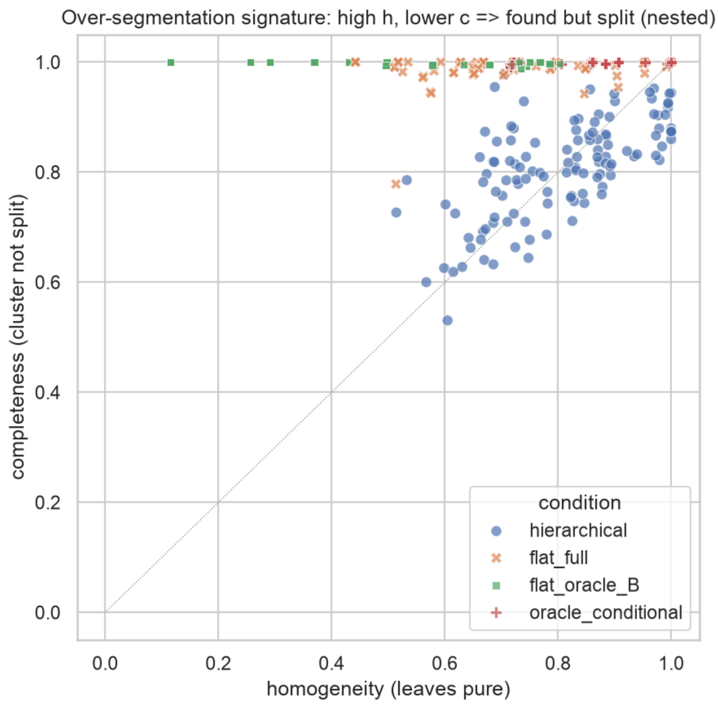


Figure 6.5.: Homogeneity versus completeness per cell (nested regime). The flat clustering (orange) sits at the top edge, merging fine sub-clusters, while the hierarchical leaves (blue) spread along the diagonal, finding the fine structure at the cost of mild over-splitting.

over each condition's own non-noise rows, and the conditions differ sharply in how much they exclude (in the nested regime the hierarchical leaves label 14.8% of rows noise on average, the flat baseline 0.1%), so the hierarchy is scored on the subset it kept and the nested deltas should be read as upper bounds. Two observations bound this artifact: the flat baseline fails by merging fine sub-clusters, which no row exclusion cures, and the non-nested control scores under the identical rule, with the hierarchy still excluding 10.8%, yet reverses the result.

Faithfulness add-on. The planted labels also enable the method-neutral faithfulness check H1a could only approximate, asking whether the leaf's local 2D view separates the planted fine labels better than the global view (by fine-label kNN agreement, per true group). For UMAP the answer is essentially no: the local−global delta is $+0.0009$ on average, drifting from -0.003 at $\rho = 1$ to $+0.005$ at $\rho = 32$. This is consistent with

H1a: UMAP leaf views gain little faithfulness. On nested-subspace data the hierarchy's benefit is therefore *partitional* but not representational. On the benchmarks it was representational (PCA/MDS) but not partitional. In both cases the hierarchy helps on the axis where the flat baseline is weak, and on neither where it is already strong.

Scope. These results fix the DR method to UMAP, the rotation diversity to a single value (so the rotation-sweep diagnostic was not run), and d_{noise} and `min_cluster_size` to single settings. Concretely, within-cluster spread is $\sigma = 0.4$ against unit-variance noise dimensions, base centroid separation is 3.0 at $\rho = 1$, and each group's fine subspace is rotated by a random rotation whose angle scales with a diversity parameter fixed at 1.0. Broader robustness sweeps were not completed, and the recovery numbers are a within-UMAP existence proof, not a method-general law. The construction is favorable to a sample-recursive method (the symmetry noted in the datasets section), so the experiment establishes that the hierarchy *can* exploit nested multi-scale subspace structure where it exists, not that such structure is common in real data.

Verdict on the hierarchy (EQ2, EQ3). Together, the two experiments yield a conditional answer. On benchmark class labels the hierarchy over-segments and recovers nothing better than a flat clustering. On planted nested-subspace structure it recovers the conditionally visible fine clusters that an unaided flat clustering merges by construction, significantly and at every scale separation (while a flat clustering given the fine subspace outright overtakes the recursion from $\rho = 4$, the feature-selection gap taken up in Future Work), with the non-nested control confirming the effect is the predicted mechanism rather than an artifact. EQ2 and EQ3 are thus answered *conditionally yes*, and the condition is specified: the recursive decomposition pays off when interesting structure is nested and multi-scale across differing subspaces, and the earlier negative is the correct result when it is not. The representational half is method-dependent (clear gains for PCA and MDS, marginal for UMAP, none for t-SNE) and matches the tuning experiment's lesson that density structure and label agreement are distinct axes.

6.8. Predicate Stability under Relaxation (EQ4)

This experiment answers EQ4. Its pre-specified hypothesis *H2* makes the question falsifiable: as the relaxation threshold is lowered from $t = 1.0$ (Section 4.8), the predicate describing a selection becomes more stable under small perturbations of that selection (higher Jaccard of the admitted sets, lower standard deviation of F1), while its ability to separate the selection (F1) degrades only gradually, so that a favorable operating point exists. One caution up front: relaxation removes tail values, and tail values are what small selection perturbations move, so *some* stability gain is built in. The experiment therefore measures what construction does not guarantee: the size of the specificity cost, whether an operating point exists on the threshold grid, and whether the severity-proportional tail split, the stated novel element of the relaxation method, outperforms a naive 50/50 split.

Setup. Selections come from the scripted simulated user (Section 5.5), logged as run `20260711_115849`. On *wine quality* (real, 11 features, 1,000-row subsample) the hierarchy is built headless (UMAP, depth two) five times, and every leaf with at least 20 members becomes a base selection (55 leaf–build units from five builds, 9–13 leaves per build; leaves are matched across builds only by member overlap, see the rebuild paragraph below, so the units are not independent). On *synthetic* data, the axis-parallel generator plants $C = 4$ clusters of 60 points, each owning a disjoint set of $r = 3$ relevant dimensions, against 15 shared noise dimensions ($D = C \cdot r + 15 = 27$), in four primary arms crossing within-cluster shape (symmetric Gaussian vs. lognormal-skewed) with box margin (wide vs. tight), plus an adversarial bimodal arm. Margins are 4.0 (wide) and 1.5 (tight) background standard deviations between the cluster center and the $\mathcal{N}(0, 1)$ background along relevant dimensions. The skewed arms use a standardized lognormal with shape 0.75 (skewness ≈ 2.9), heavy tail toward the background. Each planted cluster is a base selection (4 selections $\times$ 20 seeds $=$ 80 units per arm). A perturbed re-selection replaces a fraction δ of the selection, drawn uniformly, with points from the $k{=}10$ nearest non-selected neighbors of its members (boundary jitter, mimicking a slightly different lasso), with $m = 20$ replicates per selection, $\delta = 0.1$ as the pre-specified headline and $\{0.05, 0.2\}$ as robustness. Both predicate methods run on identical selections: the *dense* threshold predicate (one interval clause per feature) and the *sparse* DimBridge-style greedy predicate (`db`), each under both tail splits (severity-proportional vs. symmetric), over $t \in \{1.0, 0.95, 0.9, 0.8\}$. Stability is the mean pairwise Jaccard of the admitted sets over the m replicates and the standard deviation of F1, both aggregated to one value per selection *before* any test, so that replicates are not treated as independent units. Specificity is F1 of the unperturbed predicate against its base selection. The operating-point criterion was fixed in the experiment design before any run: t^* is the largest $t < 1.0$ whose median per-selection Jaccard gain over strict is positive at Holm-corrected $p < 0.05$ *and* whose median F1 stays at or above $0.9\times$ the strict median. Two empty admitted sets count as Jaccard 1, so stability without separation is possible by construction, hence the joint criterion.

F1 variance and admitted-set stability. The F1-variance half of H2 holds broadly: lowering t reduces the run-to-run variability of F1 nearly everywhere, and on wine the

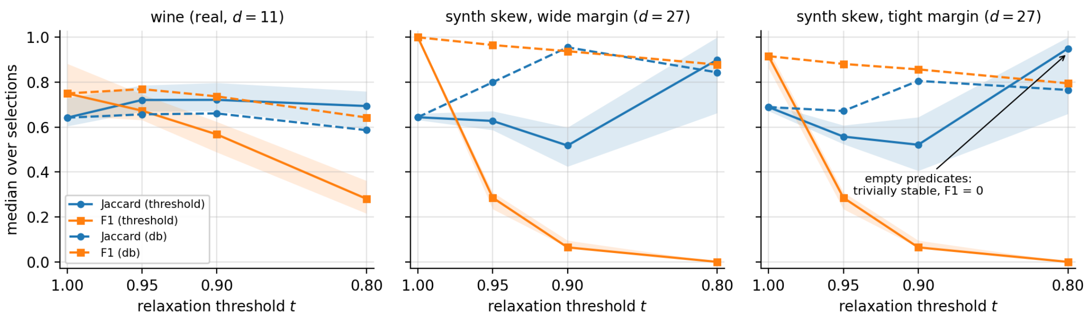


Figure 6.6.: Admitted-set stability (Jaccard, blue) and specificity (F1, orange) against the relaxation threshold t (severity split, $\delta = 0.1$; medians over selections, IQR bands for the dense method). Dense (threshold) predicates: solid; sparse (db): dashed. Panels: wine (left), wide-margin synthetic (middle), tight-margin synthetic (right). The dense $t = 0.8$ stability spike is the empty-predicate artifact.

median per-selection F1 standard deviation halves already at $t = 0.95$ ($0.064 \to 0.030$; win rate 1.00, Holm $p < 10^{-3}$; similar for db). The admitted-set half splits by data type. On wine the dense predicate's median Jaccard rises from 0.642 (strict) to 0.721 at $t = 0.95$ ($\Delta = +0.059$, win rate 0.80, $p < 10^{-4}$) and stays there at $t = 0.9$ ($\Delta = +0.038$, $p = 0.002$). These p-values rest on the per-selection unit declared above; under the conservative build-level aggregation the gain holds in all five builds (per-build medians $+0.04$ to $+0.07$), though five pairs bound the two-sided Wilcoxon at $p = .0625$. Here and throughout, Δ is the median of the paired per-selection differences, which need not equal the difference of the two reported medians. On the synthetic arms the same statistic *falls* at $t = 0.95$ and 0.9 (e.g. skew-tight: $\Delta = -0.126$ and -0.147, win rates ≤ 0.15): the strict bounding box sits in the empty margin around a planted cluster, where perturbations barely move it, whereas the trimmed bounds sit inside the cluster's dense mass, where every re-selection shifts them across many points. The apparent stability recovery at $t = 0.8$ (Jaccard 0.9–1.0) is the empty-set artifact noted in the setup: the predicates admit nothing (F1 = 0). Figure 6.6 shows all three regimes.

Why the dense predicate collapses: per-clause trims compound. The mechanism behind the synthetic collapse is arithmetic, not noise. The threshold t is applied *per clause*, and the dense predicate conjoins one clause per feature. For approximately independent features the fraction of the selection that survives all d clauses is therefore about t^d, not t. The data track this law closely, though not exactly (Figure 6.7). On the

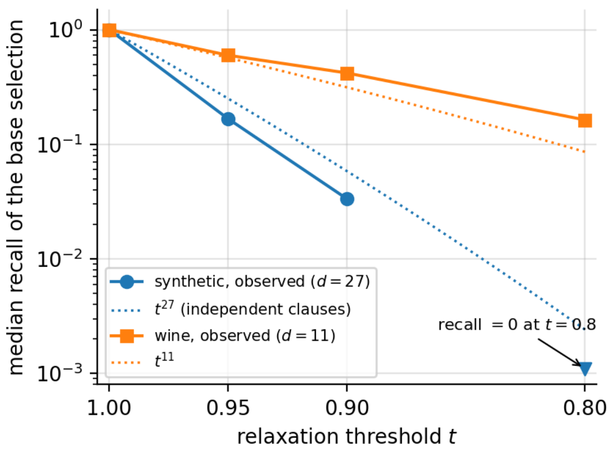


Figure 6.7.: Median recall of the base selection under the dense predicate against t, log scale, with the independent-clause prediction t^d dotted. The synthetic arms ($d = 27$) fall somewhat below t^{27}, while wine ($d = 11$) decays more slowly than t^{11} because its features are correlated.

synthetic arms ($d = 27$) the median recall of the base selection is 0.167 at $t = 0.95$, 0.033 at $t = 0.9$, and 0 at $t = 0.8$, against predicted t^{27} values of 0.250, 0.058, and 0.002: the right order of magnitude throughout, though the observed values sit somewhat below the prediction (most visibly 0.167 vs. 0.250 at $t = 0.95$). On wine ($d = 11$) the observed values sit *above* the t^{11} prediction at every threshold because correlated features trim overlapping mass. Three consequences follow. First, t cannot be read as "the fraction of the selection retained": the retained fraction is $\approx t^{d_{\mathrm{eff}}}$, where d_{eff} informally denotes the number of approximately independent active clauses ($d_{\mathrm{eff}} \approx d$ for independent features, lower under correlation, and not estimated by the method), so the same t is mild at $d_{\mathrm{eff}} \approx 5$ and catastrophic at $d_{\mathrm{eff}} \approx 27$. Second, the pre-specified grid, chosen when t was believed to act per predicate, probes far deeper trims than intended on high-dimensional data. Third, relaxation and clause *selection* are complements, not alternatives: trimming is affordable precisely when few clauses do it.

Operating points. Under the pre-specified primary analysis (dense predicate, severity split, $\delta = 0.1$), no arm satisfies the joint criterion. Wine comes closest: at $t = 0.95$ the stability gain is significant ($p < 10^{-4}$), but the F1 ratio is 0.898 against the 0.9 floor. The secondary arms locate where the favorable operating point lives. For the sparse `db` predicate (median length 1–3 clauses on the synthetic arms, so compounding is mild), the criterion is met on both wide-margin arms at $t^* = 0.95$ (ΔJaccard $+0.137$ to $+0.158$,

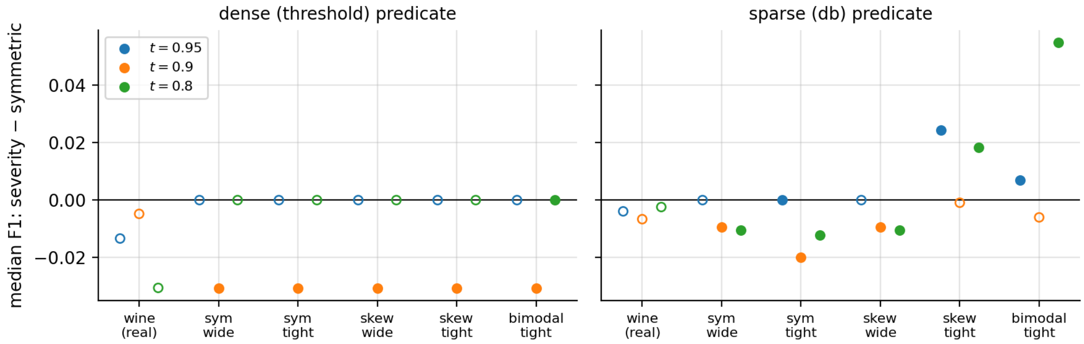


Figure 6.8.: Median F1 difference (severity $-$ symmetric split) at matched t, for the dense (left) and sparse (right) predicate. Filled markers are Wilcoxon $p < 0.05$. Negative values favor the symmetric split.

F1 ratio 0.97, robust across all three δ) and on skew-tight at $t^* = 0.9$ ($\Delta = +0.116$, F1 ratio 0.94). On wine the sparse predicate shows no stability change at $t \geq 0.9$ ($p \approx 0.7$) and no F1 cost (ratio 0.98–1.03), but its median length drops from 10 clauses (strict) to 6 at $t = 0.95$ and 5 at $t = 0.9$: relaxation buys the sparse predicate *brevity* rather than stability, a gain on the interpretability proxy (predicate length) at essentially no specificity cost.

Severity vs. symmetric split. The ablation compares the severity-proportional split against the naive symmetric split at matched t (Figure 6.8). The design predicted severity wins on skewed data and parity on symmetric data. Neither happened. For the dense predicate the severity split never beats the symmetric split anywhere. At $t = 0.9$ it is significantly *worse* on every synthetic arm, including the lognormal arms built to favor it (median F1 difference -0.031, win rates ≤ 0.05, $p < 10^{-4}$), and on wine the difference is indistinguishable from zero at every t ($|\Delta| \leq 0.03$, $p \geq 0.05$). For the sparse predicate the severity split wins narrowly on the tight skewed and bimodal arms at some thresholds ($+0.007$ to $+0.055$) and loses on the wide and symmetric arms. The failure on *symmetric* data, where the design predicted parity, identifies the mechanism: the severity of a tail is estimated from the single most extreme value on that side, the noisiest statistic a sample offers, so on symmetric data the split allocates the trim asymmetrically at random and cuts deeper into dense mass than an even split would. The severity split is thus not supported by these data, and the symmetric split is the safer default: the relaxation *mechanism* (quantile trimming) carries whatever benefit exists, while the novel allocation rule adds noise.

Table 6.6.: Predicate recovery on the axis-parallel planted clusters by relaxation threshold t (medians over the four primary synthetic arms, severity split). Dimension P/R score the recovered relevant-dimension set, membership F1 scores the admitted set against planted membership, and length is the median clause count of the sparse predicate (the dense predicate always uses all 27). Because every cell is a median over arms, columns are not mutually consistent: in particular, the dimension-F1 median is not the harmonic mean of the precision and recall medians shown.

method	t	dim. precision	dim. recall	dim. F1	membership F1	length
dense	1.0 (strict)	1.00	1.00	1.00	0.99	27
	0.95	1.00	1.00	1.00	0.29	27
	0.90	1.00	1.00	1.00	0.06	27
	0.80	1.00	1.00	1.00	0.00	27
sparse	1.0 (strict)	0.71	0.67	0.50	0.99	1–11
	0.95	1.00	0.50	0.65	0.92	1–3
	0.90	1.00	0.50	0.65	0.90	1–2
	0.80	1.00	0.33	0.50	0.84	1–2

Recovery of planted structure. Because the synthetic ground truth is known, the same runs score recovery (Table 6.6, medians over the four primary arms, severity split). The dense predicate ranks the planted relevant dimensions highest by interval selectivity at every t (perfect precision and recall at the planted count), but its membership F1 collapses with t as the compounding law dictates. The sparse predicate is the mirror image: strict greedy induction occasionally admits noise dimensions, which relaxation cleans up at the cost of dropping true ones, while membership F1 degrades gracefully. Identifying *which* dimensions matter and *describing* the members are different tasks, and each method is good at one of them.

Stability across tree rebuilds. The secondary perturbation family (rebuild the wine hierarchy with the stochastic embedding and compare predicates of matched leaves) conflates tree instability with predicate instability. Leaves are therefore matched by member overlap: mean match Jaccard 0.84 over 84 strong pairs, with the 14% of pairs matching below 0.5 excluded and counted as tree-level instability. Within strong matches the dense predicate's cross-rebuild Jaccard is 0.78 strict and 0.84 at $t = 0.95$ (mild relaxation does not hurt, and slightly helps, reproducibility across rebuilds), while the sparse predicate declines monotonically with relaxation ($0.78 \rightarrow 0.57$ at $t = 0.8$), consistent with its greedy clause selection rerouting once trimmed clauses change the

F1 landscape.

Limits. The perturbation operator (uniform member dropout refilled from boundary neighbors) is one model of "a slightly different lasso". Real selection noise is not uniform, and all stability claims are relative to this operator and to $\delta \leq 0.2$. Wine leaves within one tree build share that build, so the 55 wine units are not fully independent. The synthetic arms, whose selections are planted rather than tree-derived, carry no such dependence and agree in direction. The axis-parallel generator is favorable to range predicates (the counterpart of the EQ3 generator's bias), so the synthetic results bound what relaxation can do in the friendliest geometry. Finally, the 0.9 F1 floor in the operating-point criterion was fixed in advance but remains a judgment call: under a 0.85 floor, wine would pass at $t = 0.95$.

Verdict (EQ4). H2 as stated is refuted for the dense predicate at the tested thresholds: the stability gain exists on real data but does not coexist with acceptable specificity, because per-clause trims compound across the conjunction (Figure 6.7). Two remedies follow. Relaxation can run after clause selection (the sparse predicate meets the criterion at $t^* = 0.95$–0.9 where clusters have margin), or the trim can be budgeted per predicate rather than per clause ($t_{\text{clause}} = t^{1/d_{\text{eff}}}$). The experiment motivates the latter change but does not evaluate it. The severity-proportional split, the relaxation method's stated novel element, is not supported, and the symmetric split is the safer default. Under selection noise on wine, relaxation halves the F1 standard deviation. For sparse predicates, it also produces substantially shorter descriptions at negligible cost on wine and digits, a gain that does not transfer to breast cancer's small, high-noise leaves (Section 6.9).

6.9. Benchmark Dataset Experiments

The controlled experiments of Sections 6.5–6.8 establish *where* the hierarchy and the relaxation pay off. This section reports what the *shipped defaults* produce on three real benchmarks from Table 6.1 (wine quality, breast cancer, digits), where no ground-truth subspace exists. To keep a descriptive section from post-hoc storytelling, every reporting and aggregation rule, including the rule that picks the example predicate of Table 6.7, was fixed in the experiment design before the run, and four pre-specified *consistency checks* (C1–C4), out-of-sample predictions derived from the completed experiments, give the section falsifiable content. A failed check is a finding, reported with the same prominence as the H1b and H2 negatives.

Table 6.7.: Shipped-default behavior per dataset. Cells are medians (min–max) over five unseeded builds. Mean trustworthiness/continuity are unweighted means over a build's leaves. The last column is the fixed-rule example leaf (median-leaf-count build, median-size leaf): sparse `db` clause count and F1, strict → relaxed at $t = 0.95$. †: a branch hit the depth cap of four. Leaves do not partition the data: the per-build noise fraction (points dropped as HDBSCAN noise mid-recursion, reaching no leaf) is 0.26 (0.20–0.32) on wine, 0.43 (0.32–0.59) on breast cancer, and 0.10 (0.09–0.18) on digits.

Dataset	depth	# leaves	mean trust. / cont.	example leaf predicate
Wine quality	3 (3–4†)	16 (15–17)	0.81 (0.80–0.83) / 0.83 (0.83–0.85)	n=42: 10 → 8 clauses, F1 0.61 → 0.64
Breast cancer	3 (3–4†)	7 (6–9)	0.78 (0.76–0.79) / 0.81 (0.76–0.81)	n=27: 8 → 6 clauses, F1 1.00 → 0.83
Digits	3 (2–3)	16 (15–17)	0.83 (0.83–0.84) / 0.85 (0.84–0.86)	n=53: 12 → 5 clauses, F1 0.97 → 0.92

Setup. Datasets are prepared as in the other harnesses (seeded 1,000-row subsample; breast cancer, at 569 rows, in full) and run through the unmodified pipeline driver under the shipped defaults: UMAP,[1] root z-score standardization, pre-clustering reduction to two components, HDBSCAN at minimum cluster size 25 and minimum samples 5 (run `20260728_185329`, depth-2 diagnostic `20260728_190741_depth2_diag`). The tuning study of Section 6.5 scored this same configuration (on its 500-row subsamples) at DBCV −0.40, −0.21, and −0.70 on wine, digits, and breast cancer: the walk probes the shipped operating point, not a tuned one. The one named deviation is a depth cap of four instead of the harness convention of two, so realized depth is *discovered* by the stopping rules of Section 4.3, and a branch that still hits the cap is marked †. Five builds per dataset act as replicates over the unseeded UMAP stages (a degenerate build would be reported, never re-rolled; none occurred). On every leaf with at least 20 members, both predicate methods run at $t \in \{1.0, 0.95, 0.9, 0.8\}$ under the severity split, the shipped default, kept although Section 6.8 recommends the symmetric split. Everything here is within UMAP, the dominant axis of variation (Table 6.3), so nothing in this section generalizes across DR methods.

Shape and faithfulness. Table 6.7 carries the shape numbers. The design's prediction that the depth cap of four would never bind was wrong in three of the ten wine and

[1]The walk ran under the then-current calc-layer default exploration method (UMAP). At the released tag the calc-layer default is PCA and the walk's driver inherits it. The node embeddings reported here are UMAP.

breast-cancer builds, which the † entries record. The noise fraction is the number the leaf count hides (caption of Table 6.7). C3 below returns to it. Leaf-local faithfulness is high in absolute terms but sits below the root embeddings' global scores (0.91–0.96), a soft comparison, since leaf scores mix neighborhood sizes $k = 12$–20 where the root always scores at $k = 20$. The k-matched arm C2 below is the inference-grade contrast, and it points the other way.

Predicate readability. The sparse `db` predicate stays short even strict: pooled median lengths are 10 (wine), 9 (breast cancer) and 12 (digits) clauses against $d = 11, 30, 64$: on high-dimensional data the greedy clause selection, not the relaxation, does most of the compression. Relaxing to $t = 0.95$ shortens the medians to 6, 7, 7 (C4 below) at a specificity cost that varies: median F1 moves $0.77 \to 0.74$ on wine and $0.94 \to 0.89$ on digits, but $0.91 \to 0.78$ on breast cancer. The finding of Section 6.8 that sparse relaxation is nearly free on wine does not transfer to breast cancer's small, high-noise leaves. The example column of Table 6.7 carries the fixed-rule example leaf per dataset, and a fully worked strict-versus-relaxed reading on wine, in original units, is the predicate step of the case study (Figure 6.11, Section 6.10).

Consistency with the controlled experiments. Each check gets its prediction, observation, and verdict: C1 and C4 pass, C3 fails on two of three datasets, and C2 fails narrowly.

C1: compounding, out of sample. Section 6.8 established dense-predicate recall $\approx t^d$ at $d = 11$ and 27. Breast cancer ($d = 30$) and digits ($d = 64$) extend the curve to dimensionalities never fit. Predicted: median dense recall of leaf members at $t = 0.95$ at or above the independence prediction t^d yet far below one, with digits below 0.5. Observed: 0.34 against the predicted 0.21 (breast cancer) and 0.38 against 0.04 (digits), with wine in-sample at $1.01\times$ its prediction. **Pass**, extending the compounding pattern of Figure 6.7 to $d = 30$ and $d = 64$. Digits sits an order of magnitude above its prediction because correlated pixel features trim overlapping mass, the mechanism that put wine above t^{11}, yet a single relaxation step still costs 62% of the leaf's members, which is the compounding law's point.

C2: the H1a replication arm. For every leaf with $n \geq 41$ (so $k = 20$ on both sides), the leaf's own embedding is scored against the root embedding sliced to the same rows: the exact H1a contrast, predicting a small positive median Δtrustworthiness (at most 0.05) with a win rate in $[0.5, 0.8]$. Observed, pooled over 101 eligible leaves: median $+0.021$ (continuity $+0.007$), Wilcoxon $p \approx 10^{-11}$ (the section's only significance test, computed over build-sharing leaves and therefore anti-conservative), but a win rate of 0.81 (0.85, 0.88, 0.76 on wine, breast cancer, digits). The magnitude replicates

H1a. The consistency does not: **fail**, by 0.01 on the pre-specified ceiling. The local re-embedding advantage is more uniform across leaves here than H1a's "roughly a wash" characterization predicted.

C3: noise fractions. H1b's UMAP cells, on the same datasets and defaults, predict per-build noise fractions of 0.00–0.19 (wine), 0.15–0.32 (breast cancer) and 0.12–0.17 (digits). Observed: 0.20–0.32, 0.32–0.59 and 0.09–0.18. The decision rule fixed before execution treated modest excursions as expected consequences of the different depth cap and factor-level disagreement as failure. Digits straddles its predicted range on both ends by at most 0.03 (**pass with excursion**: outside the point prediction, within a modest excursion). Wine and breast cancer lie substantially above their predicted ranges (**fail**). A depth-two rerun of the identical harness path returns wine to 0.04–0.16 (inside the H1b range) and digits to 0.09–0.16, attributing wine's excess to noise compounding across the two extra recursion levels the cap of four allows. Breast cancer's depth-two values (0.10, 0.11, 0.51) additionally straddle the H1b range on both sides, exposing build-to-build variability on 569 rows that H1b's five builds understated. No harness defect is indicated (the code path is identical), but the number is a tool-experience fact: under shipped defaults with deep recursion enabled, the breast-cancer tree's leaves cover a median 57% of the data.

C4: sparse brevity. Predicted: wine replicates the strict-to-relaxed median length drop $10 \rightarrow 6$ of Section 6.8 within ± 2 clauses. Breast cancer and digits show the same direction with strict median $\ll d$. Observed: wine $10 \rightarrow 6$ exactly, breast cancer $9 \rightarrow 7$, digits $12 \rightarrow 7$. **Pass** throughout.

Purity (secondary). Leaves are typically class-pure, with median purity 1.00, 1.00, and 0.98 and enrichment 1.3, 1.6, and 9.6. However, classes are not clusters [Jeo+24] and purity is not the objective. The pre-specified expectation, from the tuning experiment's $r \approx 0.17$, was a weak faithfulness–purity association. Observed Spearman ρ is $+0.06$ (wine), -0.10 (breast cancer), and -0.38 (digits), which is weak on two datasets. The stronger negative association for digits is driven by the few impure leaves. No significance test was performed for this secondary analysis.

Only C1 and C4's direction on breast cancer and digits are genuinely out of sample. C2 and C3 reuse H1a/H1b's datasets and defaults and test internal coherence. The shipped pipeline behaves on real benchmarks largely as the controlled experiments predict, and where it does not (C2's uniformity, C3's depth-driven noise), the discrepancies are mechanistically attributable and are findings in their own right.

6.10. Case Study and Expert Feedback

Two qualitative components complement the quantitative experiments: a case study, reported here as two walk-throughs through the prototype (digits as the main narrative, wine for the predicate step), and two moderated expert sessions built around the two walk-throughs. In the terminology of design-study methodology [SMM12], the walk-through is a usage scenario, an author-driven demonstration on benchmark data rather than an evaluation with an external analyst, and its evidence is read in that limited sense: it illustrates on real data what the experiments of this chapter measure, and it claims nothing a demonstration cannot show. Following the nested-model discipline [Mun09], each observation below is tied to the level it belongs to, the data and task abstraction or the concrete views, and is not cited as evidence for any other level.

Stimulus selection. The dataset and configuration were fixed by a screening sweep, not by choosing a flattering example after the fact. Every dataset that could be loaded in the screening environment (the six of Table 6.8; of the ten-loader registry, Olivetti faces and Fashion-MNIST could not be downloaded in the screening environment, MNIST's raw files were absent from it, and QM9 was not attempted) was built under both shipped configuration variants (the calc-layer defaults, which pre-reduce to two dimensions with UMAP, unseeded at the tag under which the sweep ran (see below), and the frontend defaults, which cluster in the full feature space and are deterministic) at recursion depths one to three, with five repeated builds wherever the pipeline was unseeded. Table 6.8 reports the depth-two rows. Two facts drove the choice. First, on wine the pre-reducing variant reaches depth two only by shattering the data into dozens of leaves (Table 6.8), with no two of five builds agreeing, which is unusable as a fixed stimulus. Second, the full-space variant on wine is deterministic and produces the smallest non-trivial hierarchy of the sweep: two top-level regions and four leaves. The case study therefore uses wine quality with the frontend defaults and three documented changes: recursion depth two instead of the shipped depth one (a depth-one tree has no second drill-down level to demonstrate), the PCA exploration view (the calc-layer default, while the app opens on UMAP), and the local (this-cluster) predicate scope (the app opens on the global scope). Wine is also the only screened dataset whose features carry meaningful names, a prerequisite for reading range predicates at all, and the real dataset of the EQ4 experiment (Section 6.8), so the qualitative and quantitative evidence refer to the same data. The sweep also shows how often the shipped pipeline finds multi-level structure on real data at all: under the deterministic variant, two of the four real datasets reach depth two (wine and digits, both discarding over half their points on the way). Under the pre-reducing variant every dataset reaches depth two,

Table 6.8.: Screening sweep at recursion depth two: leaves, top-level regions, and the fraction of points reaching no leaf, per dataset and configuration variant. Calc-layer default ranges span five unseeded builds. Frontend-default builds are deterministic (repeats byte-identical). Under the frontend defaults iris stops at depth one and breast cancer does not split at all. On the two-dimensional rings dataset the variants coincide, as pre-reduction is inactive there.

	calc-layer defaults (2-D UMAP)			frontend defaults (full space)		
Dataset	leaves	regions	no leaf	leaves	regions	no leaf
Wine	69–92	9–64	.21–.34	4	2	.58
Iris	3–4	2	.03–.06	2	2	.01
Breast c.	5–6	3	.14–.32	1	0	.00
Digits	21–27	12–16	.13–.19	9	8	.56
Rings	15	3	.13	15	3	.13
Swiss roll	21–25	7–12	.10–.16	12	10	.33

but, wherever pre-reduction is active, never twice in the same shape.

Second stimulus: digits, found interactively and then disciplined. The digits stimulus has a different provenance, disclosed in full. Exploring digits in the interface with UMAP pre-reduction to 32 components, a setting between the two shipped variants, surfaced a region mixing fours and sevens. Two problems had to be resolved before that observation could carry weight. First, the build was unrepeatable: the interface displays a random state for UMAP, but the frozen code never passed it to the reducer, so the region was one arbitrary draw. The config's random states are now threaded into the reducers. That, the chunked neighbor-metric path of Section 5.3, and a deterministic PCA solver choice are the code changes since the interface freeze, and the tag was moved accordingly. Second, one build is an anecdote. A seed sweep (five seeds, depths one and two, all else fixed) shows: at depth one, two of five builds contain a mixed-class cluster (one mixing fours, sevens and nines; one mixing fives and nines); at depth two, no mixed cluster survives in any of the five builds, because the second level separates the classes the first level merges. Seed 4 reproduces the interactively found region almost exactly and is fixed as the stimulus: digits, depth two, pre-reduction to 32 components, UMAP views, seed 4, deterministic on a fixed machine and environment.

First walk-through: digits, a confusable region and its resolution. The build reads all 1,797 digits with the 64 pixel features. The ten one-hot label columns are withheld and surface only as target characteristics. The root overview shows eleven regions with 1.8% noise (trustworthiness 0.972, stress 0.583). Region C2 ($n = 386$) is the interesting one: its characteristics bars on the withheld labels flag a mixture, 46% sevens, 46% fours, 8% nines in the bars' whole-number rounding (Figure 6.9, left). Drilling into C2 (at 8.5% internal noise the lowest of any region the build recurses into) resolves it: C2 splits into C2.0, 171 points that are all fours, and C2.1, 182 points that are 97% sevens. The 33 points the split refuses to assign are 88% nines, and the row-image view shows why (Figure 6.9, bottom): they are the instances a human would also hesitate on, drawn between classes, and they sit visibly between the two lobes of the re-projection (Figure 6.9, right). Two additional analyses constrain this interpretation. First, the confusion is not intrinsic to the data: within C2 in the original 64-pixel space, only 4.9% of points have even one differently-labeled neighbor among their ten nearest, so the merge is introduced by the 32-dimensional reduction, and the local step recovers the separation. That is a conditionally visible structure of the same shape as EQ3's (Section 6.7), though the concealment here is introduced by the pipeline's own 32-component reduction rather than by the data: structure invisible at one level, recovered by recursion. Second, the local UMAP views do not gain faithfulness along this path (trustworthiness 0.940 at C2, 0.903 and 0.910 at its leaves), consistent with EQ2's finding that the local-view advantage is marginal for UMAP. The wine walk-through below, under the PCA view, shows the opposite behavior. Total attrition of this build is 20.8%.

Second walk-through: wine. The build reads all 6,497 wines with the eleven physicochemical features. The two label columns (color and quality score) are withheld from the feature space and surface only as target characteristics. The root overview (Figure 6.10, left) shows two regions and the noise points the density clustering assigns to neither, 34% of the dataset at this level. The faithfulness tiles for this global view read trustworthiness 0.851, continuity 0.959, stress 0.375. Checking the two regions against the withheld color label is the walk-through's verification moment: region C0 ($n = 3470$) is 99.8% white and region C1 ($n = 832$) is 99.9% red. The tool never saw the color, and the split emerged from the chemistry alone. Drilling into C0 re-runs the analysis on its points, and the local view (Figure 6.10, right) reports higher faithfulness on its own tiles, trustworthiness 0.876 and stress 0.317, rising further at the leaf below (trustworthiness 0.955, stress 0.245). Each node's tiles are computed over that node's own points, so these cross-level readings are indicative rather than the controlled same-rows comparison of Section 6.6; the rise along the drill-down path echoes the EQ2 local-view advantage of H1a, observed here with the PCA exploration view (one of the three documented changes). The re-projection separates a compact lobe, C0.1 ($n = 39$), from the bulk white leaf C0.0 ($n = 2523$): its characteristics identify a sweet-white

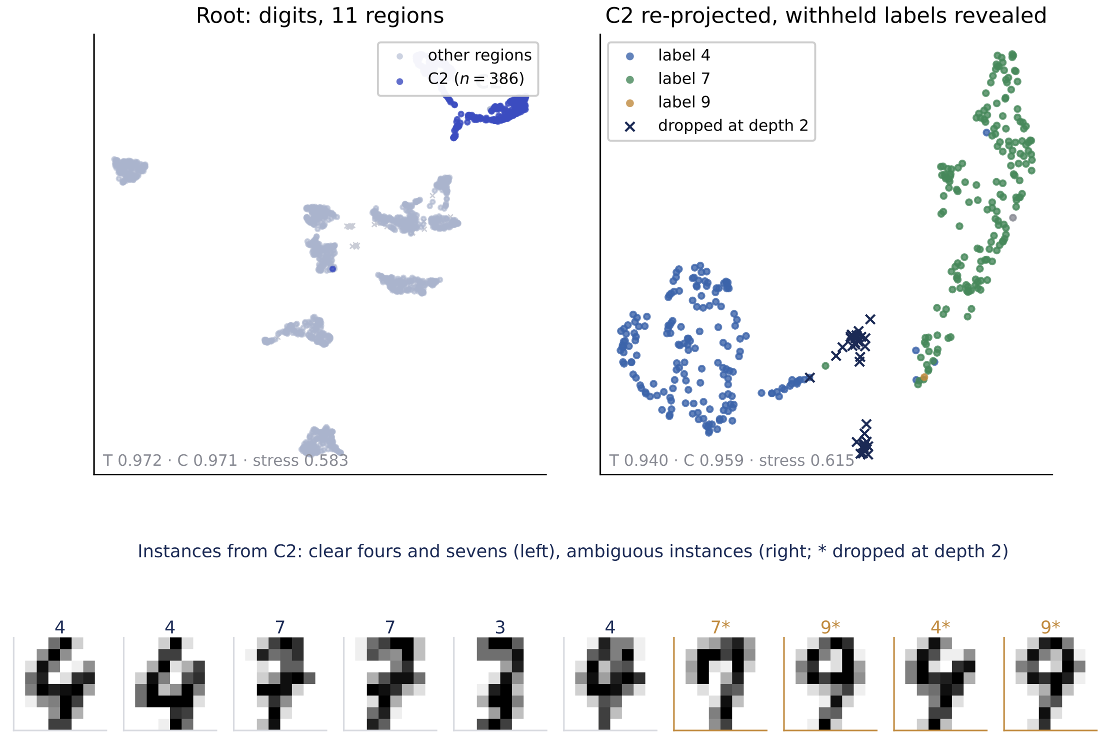


Figure 6.9.: The digits walk-through. Left: root overview, region C2 highlighted. Right: C2 re-projected on its own points with the withheld labels revealed. Fours and sevens separate, and the points dropped at depth two (crosses, 88% nines) lie between the lobes. Bottom: instances from C2 via the row-image view. Clear fours and sevens, one stray three that the rounded composition conceals, then ambiguous instances (starred: dropped at depth two).

group, median residual sugar 14.6 versus 5.2 g/l in C0, density 0.9995 versus 0.9936, alcohol 9.0 versus 10.4 vol%. The withheld quality score barely moves (mean 6.03 versus 5.94), a useful negative: the discovered chemistry does not simply re-encode the label.

Selection and predicates. Lasso-selecting C0.1 within region C0 and generating its description under the local (this-cluster) predicate scope yields the strict and relaxed predicates of Figure 6.11. The strict predicate needs seven clauses (F1 0.940 against the selection). The relaxed predicate at $t = 0.9$ needs three, sulphates in $[0.67, 0.82]$, citric acid in $[0.33, 0.44]$ and alcohol in $[8.6, 9.5]$, at F1 0.917. This reproduces the sparse-predicate brevity result from EQ4: fewer than half as many clauses for an F1 decrease of 0.023. Neither description includes residual sugar. The strict predicate carries density,

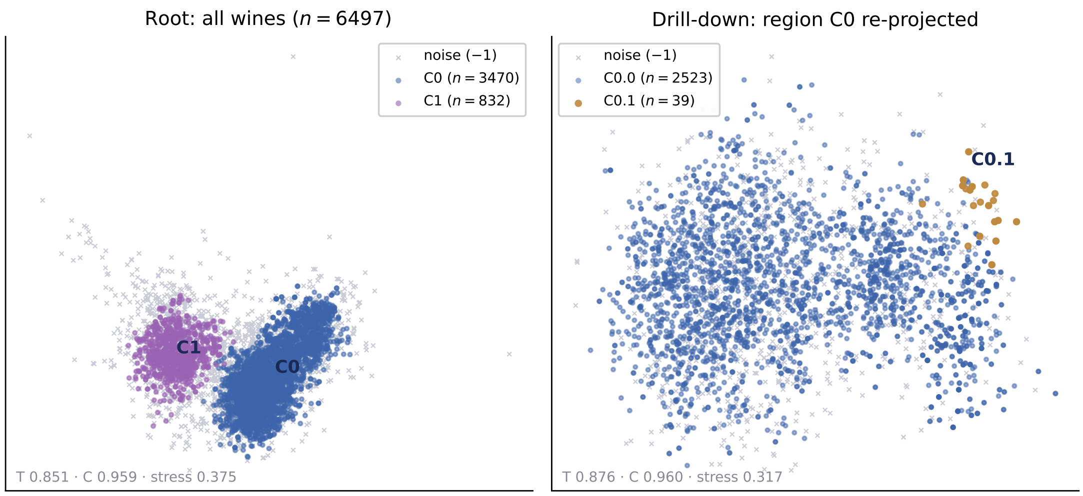


Figure 6.10.: The case-study walk-through. Left: root cluster-projection overview of all wines (two regions, noise as gray crosses). Right: region C0 re-projected on its own points. The local view reports higher trustworthiness on its own points (0.876 vs. 0.851 at the root) and separates the sweet-white lobe C0.1 that anchors the selection step.

sugar's close physical proxy, and the relaxed predicate identifies the group entirely by sulphates, low alcohol, and citric acid, discriminative within white wines where high sugar alone is not. Whether such proxy descriptions read as trustworthy to a user is precisely the judgment this walk-through cannot supply and the expert sessions are for.

Attrition. Each configuration has its own attrition. The wine walk-through discards 34% of points at the root and 58% by the leaf level, and its red branch makes the cost concrete: 77% of C1's points re-drop to noise at depth two, leaving two small pure leaves ($n = 30$ and $n = 159$), so an analyst drilling into C1 sees mostly gray. The digits build is milder (20.8% overall), and there the dropped points are informative rather than lost: C2's residue is the ambiguous instances. These are the recursive-attrition dynamics of Section 6.9 and Chapter 7 under each stimulus's own configuration, visible in the interface exactly where they occur.

Limits of the walk-throughs. The wine walk-through illustrates H1a on real data. The digits walk-through shows recovery with the same structure as EQ3, although the pipeline's own reduction introduces the concealment rather than the data. The screening sweep bounds the practical-prevalence question. These components cannot

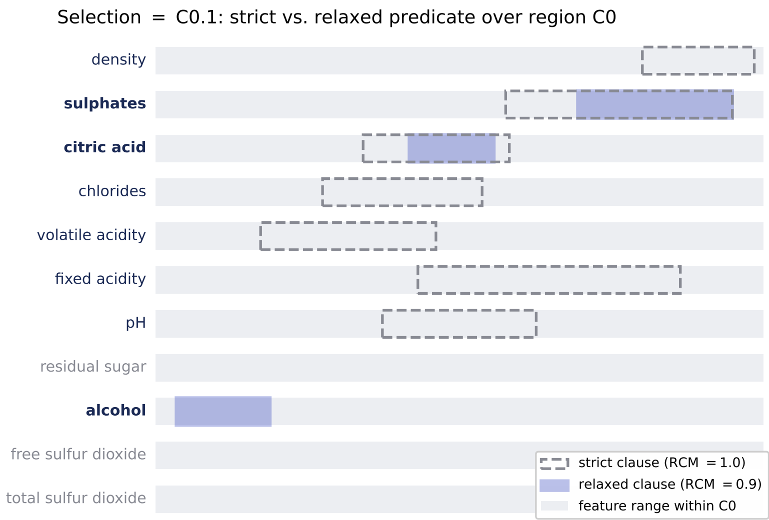


Figure 6.11.: Strict (dashed, RCM = 1.0, seven clauses) and relaxed (indigo, RCM = 0.9, three clauses) predicate for the selection C0.1, in original units, each band drawn within its feature's range over region C0. Neither names residual sugar. Density (strict predicate only) and sulphates (both) act as the sweet group's discriminators.

establish whether users find relaxed predicates easier to read and trust. The expert sessions examine that question.

Expert feedback: design. Each expert session is moderated, conducted in person, and runs 60 minutes on the frozen build, following a fixed script. The session begins with background elicitation and a digits warm-up. Participants first explore freely and then focus on region C2. The moderator directs them only if they have not entered C2 unaided, and the path taken (self-found or directed) is logged and reported. Participants then complete the scripted wine walk-through and three hands-on wine tasks: find a subgroup, describe a selection, and perturb the selection. The session ends with a questionnaire that presents the strict and relaxed predicate variants in a blind A/B comparison, followed by a semi-structured debrief. The questionnaire is organized in seven blocks, Background, The problem, The warm-up dataset (handwritten digits), Finding structure (the wine dataset), Reading the interface, The two descriptions (the wine dataset), and Value and adoption, with agreement items rated from 1 (strongly disagree) to 5 (strongly agree) and an explicit cannot-judge option. The background familiarity items use a 1 (none) to 5 (expert) scale. The first session used questionnaire v1.2, the second the minor revision v1.3 with identical items. Two administrative changes in the second session matter for reading its results: the two descriptions of the blind A/B block were prepared before the session, on a fixed selection of 39 points in the white-wine region chosen to resemble the walk-through's sweet-white selection, rather than generated on the participant's own points (a documented deviation from the protocol), and each description card states how many of the 39 points it covers, so the coverage cost of relaxation was visible during the comparison. Two sessions were conducted in August 2026, with two different participants; they are reported in turn.

The first session. The first was with a data scientist with four to ten years of experience in data analysis who works with tabular data of more than ten features weekly (self-rated familiarity, on the questionnaire's five-point scale: dimensionality reduction 4, clustering 4, reading projected scatter plots 5) and who agreed that the task the tool addresses occurs in their own work (A1: 4). In the warm-up the participant entered the mixed region C2 unaided during free exploration. Descending, they read the group from the tool's displays rather than the row images (F3: 1, negatively keyed; F2: 4 for the descent clarifying the composition) and afterwards judged that the displays alone would have sufficed (F1: 5), a stated judgment that the bound below deliberately does not promote to demonstrated sufficiency. The gray points left them undecided (F4: 3). All three wine tasks were completed, with occasional tier-one prompts ("what are you looking for?") and no substantive help.

Questionnaire and debrief, first session. The two negatively keyed interface items are the clearest results: orientation was never lost during drill-down (B3: 1) and the interface did not show more than the participant could use (C4: 1). Drill-down showed structure not visible in the global view (B1: 4) over regions judged meaningful (B2: 4), the per-region view sufficed to decide where to go next (B4: 5), and the overview projection and range chart read easily (C1: 5, C3: 5). Selection behaved as expected (item C2: 4). The open item asking for the least clear view was left blank. Asked for a surprising moment, the participant named the walk-through's verification moment and, unprompted, the brevity finding of EQ4: "surprised by the clear separation of red and white wines ... also interesting to see that some spaces can be described by only 3 or 4 of the dimensions." Asked whether the lack of an image-like check on ordinary data would change their trust in the tool's groups (F5): "not really, can still rely on the predicates and characteristics provided by the tool." This is a stated judgment, not demonstrated reliance. The participant identified performance as the strongest concern ("takes some time to load, computationally intensive ... but not too bad, would still try it") and named "faster computation, great handling of very large datasets" as the change that would most increase the tool's value (E3, E4). The participant would try the tool on their own data (E1: 5, E2: 4 for saving effort over their current approach) but could not name a concrete dataset.

The second session. The second participant is a data scientist with one to three years of experience who also works with tabular data of more than ten features weekly (familiarity: dimensionality reduction 3, clustering 4, reading projected scatter plots 3). Unlike the first, this participant disagreed that the task the tool addresses occurs in their own work (A1: 2), naming k-means followed by inspecting the raw data as their approach the last time something similar came up, so their answers are read here as a newcomer's view of the interface rather than as expert practice. In the warm-up they too entered the mixed region C2 unaided. Exploring further, they called out, unprompted, that digits of different classes were split across several regions, and explained it by those digits looking alike: the fragmentation the partitional half of EQ2 measures as over-segmentation was visible to this participant but read as visual similarity, not as an error. The hands-on wine part ran as free selection rather than as the three scripted tasks: the participant selected several point sets, generated their descriptions, and read the characteristics, again with occasional tier-one prompts and no substantive help. That the descriptions tracked their changing selections the way they expected was the moment this participant singled out as interesting.

Questionnaire and debrief, second session. The second questionnaire produced broadly similar responses, with two differences. The negatively keyed items again fall on the favorable side: orientation was not lost during drill-down (B3: 2) and the interface did not show more than the participant could use (C4: 1). Descending a level made the warm-up group's composition clearer than the top-level view alone (F2: 5), drill-down showed structure the global view did not (B1: 4) over regions judged meaningful (B2: 5), and the per-region view sufficed to decide where to go next (B4: 4). The displays would have let them judge the group without the row images (F1: 4), though this participant was neutral on whether they in fact relied on the displays rather than the images (F3: 3). This response therefore does not establish that the displays alone were sufficient. The first difference concerns the unassigned points. Unlike the first participant, this one agreed that some points the tool set aside as unassigned looked like they belonged to one of the groups (F4: 4), the first external challenge to the noise decisions whose cost this chapter quantifies throughout. The interface items sit one point lower than the first session's (C1: 4, C3: 4, selection behaving as expected C2: 3), and the least-clear-view item, blank in the first session, here names the predicate terminology: "not knowing predicates I was confused by ranges and what was meant by ‘features used’.” The second difference concerns the missing image check (F5): where the first participant said their trust would not really change, this one judged that it “maybe even would increase the trust, or let’s call it reliance,” on the tool’s other displays, because without a picture per row those displays are the easier way to get an overview. Asked for a surprising moment, they named the clear separation of the digits rather than anything on wine (B5).

The blind description comparison. The A/B block of the questionnaire was completed by one of the two participants, the second, so the planned counterbalancing of which variant appears as Description A does not apply. The participant saw the two pre-generated cards for the fixed 39-point selection described above, labeled only A and B. Unblinded here: Description A was the relaxed predicate at $t = 0.9$, five clauses covering 31 of the 39 points, and Description B the strict predicate at $t = 1.0$, ten clauses covering all 39 by construction. The participant rated the relaxed description easier to read and understand (D1: 5 for the relaxed against D2: 2 for the strict) and easier to pass on, judging that they could tell a colleague which points are meant from that description alone (D3: 5 against D4: 2). Trust that a description would still apply if similar new points were added was rated 5 for both (D5, D6). Asked which single description they would report to a colleague, the participant chose the relaxed one (D7), leaving the written reason blank but reasoning in the debrief that once one is already willing to lose information by applying a dimensionality reduction, losing a few points is worth a description that is easier to understand. The preference was therefore stated with the coverage cost in plain view, the cards disclosing 31 against 39 covered points, and it echoes the EQ4 sparse-brevity finding from the user side.

Scope of the expert evidence. Four bounds apply to the expert evidence. First, the sample is two self-selected participants: everything here is indicative, a record of how two practitioners read the tool, not a statistical result, and the description comparison rests on one of the two. Second, on digits the row images were freely available and the warm-up preceded any instruction, so digits observations count at the encoding and interaction level, plus explicitly bounded qualitative color at the abstraction level. They cannot show that the tool’s own displays would have sufficed for the structural judgment. Third, by the wine part each participant had seen drill-down twice (warm-up and demo), so the wine observations speak to guided navigation on unfamiliar features, not to unaided discovery. Fourth, the second participant stands one step further from the target task, with one to three years of experience and no such task in their own work, so agreement between the two sessions shows the reading survives a change of experience level, not that it generalizes beyond two people.

All four evaluation questions are answered by completed experiments. The verdicts are restated question by question in Section 8.2. The benchmark walk and the case study add descriptive and qualitative readings, and the expert sessions the only external ones, each within its stated bounds.

7. Discussion

This chapter interprets the experimental results, then states the trade-offs the approach accepts, its limitations, and the threats to the validity of its conclusions.

7.1. Interpretation of Results

This section reads the completed experiments back against the evaluation questions of Section 6.1, taking the hierarchy first (EQ2, EQ3), then the relaxation (EQ4), and the configuration question (EQ1) last.

For EQ2, the hierarchical-versus-flat experiment (Section 6.6) gives a nuanced answer: the decomposition helps *representationally* but not *partitionally*. A region's own projection is more faithful than the global map for PCA and MDS, marginally for UMAP, and not for t-SNE. Partitionally the experiment is a clean negative: the hierarchy over-segments and recovers ground-truth labels *worse* than a single flat clustering, with the embedding's documented false-tear tendency a candidate mechanism [MHM18b]. The negative is, however, a verdict on the measure rather than on the decomposition: benchmark class labels are coarse partitions that encode no nested subspace structure, so a flat clustering that already resolves them leaves the recursion nothing to add (Section 6.7). Where the ground truth does encode nested structure, the same ARI-based comparison reverses. The aims of Section 4.1 do not include computing clusters or an interestingness score. The first asks for a lens that makes locally relevant structure reachable and inspectable. The hierarchy's contribution is therefore the locally adapted *view* (and, through it, the region and predicate the analyst reaches), not a better cluster assignment.

The planted-subspace experiment (EQ3, Section 6.7) supplies what class labels cannot measure: when structure is nested and multi-scale across differing subspaces, recursion recovers the fine clusters a flat clustering merges by construction, and a non-nested control confirms the advantage reverses when the mechanism does not apply. The hierarchy question is therefore *conditionally yes*, not a straight rejection. The margin left to a subspace-aware oracle grows with the coarse scale because the recursion partitions samples but cannot *select* features per node (Section 6.7). The node splitter this points to is a per-node feature-weighting split in the manner of projected clustering

(PROCLUS [Agg+99]), with subspace clustering (SUBCLU [KKK04]) as the alternative enumeration strategy inside the recursion. It is taken up in Future Work.

EQ4 asked whether relaxation buys stability at an acceptable cost in specificity, and at which threshold the trade turns unfavorable. The stability experiment (Section 6.8) answers it, though not as hypothesized. The stability half holds in full only on the real dataset: on wine, relaxation halves the standard deviation of predicate quality under selection noise and the dense predicate's admitted sets stabilize, whereas on the synthetic arms the admitted sets become *less* stable ($\Delta = -0.126$ at $t = 0.95$, skew-tight), because a strict box sitting in an empty margin is already insensitive to boundary jitter. Outside the real-data regime both halves of H2 fail, not only the cost half. The cost half was misjudged at design time: because the threshold acts per clause, a dense conjunction retains roughly $t^{d_{\mathrm{eff}}}$ of the selection rather than t, and the trade turns unfavorable almost immediately in high dimensions. A dilution failure, clauses widening until the predicate admits most of the data, is the intuitive worry for a relaxation but cannot arise under this operator, which only trims clauses inward (Section 4.8). The failure mode that does arise is the opposite: over-tightening through compounding. The severity-proportional split, the method's novel element, is a reported negative: its severity estimate rests on the single most extreme value per tail, and the data prefer the naive symmetric split. That relaxation of the *sparse* predicate markedly shortens the wine descriptions at essentially no F1 cost (Section 6.8) is the finding most directly relevant to interpretability, the property the predicate descriptions are for. The case study (Section 6.10) shows the same brevity gain enacted on a live selection and its screening sweep addresses a precondition of EQ3's prevalence question, how often the shipped pipeline reaches a second level at all: two of four real datasets do under the deterministic full-space variant. Whether the structure found there is nested-subspace structure is untested. The blind comparison in the expert sessions provides limited supporting evidence: the participant who completed it rated the relaxed description easier to read than the strict one and chose it for reporting with its lower coverage disclosed. This is one observation rather than a validation.

From the tuning-effectiveness experiment (EQ1, Section 6.5), the practical reading is to reserve intelligent search for the clustering configuration and to treat DR-faithfulness tuning as optional: the density-clustering stage benefits substantially from tuning in the observed in-sample comparison, while the two-dimensional embedding is near-ceiling out of the box. Most importantly, the weak DBCV-gain–ARI coupling ($r \approx 0.17$) cautions against reading any single internal index as “quality”, which is why the evaluation treats density structure and label agreement as separate axes.

7.2. Trade-Offs

The approach exposes several deliberate trade-offs, each traceable to a design decision in the prototype. The relaxation threshold t trades the specificity of a predicate against its stability: lowering t trims outlier-driven width, narrowing the accepted region while steadying the description. Trimmed too far, however, it destroys the recall that lets the predicate cover the selection at all. This is the same precision-versus-recall tension seen from the predicate's side, and it is why the evaluation sweeps t rather than fixing a value.

The optional UMAP pre-reduction trades cleaner density structure for the clusterer against faithfulness to the original space: reducing before clustering concentrates density and helps HDBSCAN, but a cluster found in UMAP space need not be a cluster in the original space, the risk the per-node faithfulness scores of Section 4.6 expose.

A further trade-off is recursive noise attrition. HDBSCAN marks low-density points as noise at every level, and a point dropped once never reaches a leaf, so attrition compounds with depth: under the shipped defaults a median 26% of wine, 43% of breast-cancer, and 10% of digits rows reach no leaf, and the depth-two rerun of Section 6.9 attributes wine's excess to noise compounding across the extra recursion levels. The statistical consequence deserves stating: a leaf is not simply a small subset of the data but a subset conditioned on having survived every earlier noise decision, so leaf-level distributions, purity, predicate characteristics, and stability describe this selected population rather than the dataset as a whole. A noise-aware stopping or reassignment rule is taken up in Future Work.

The choice of axis-parallel range predicates trades interpretability against completeness: a conjunction of feature ranges is immediately readable, but it cannot express correlations or oblique boundaries, so a diagonal structure is described only coarsely. Finally, the quality-guided configuration search trades automation against user control: tuning frees the analyst from manual parameter choice, but the workflow is fundamentally interactive, since the analyst still chooses where to drill and what to select.

This work implements a single-path approximation of the greedy induction that DimBridge adopts from PIXAL [Mon+22] (DimBridge itself retains multiple overlapping candidate predicates), so the strict baseline follows one of the two algorithms DimBridge considers. DimBridge's own novel predicate-regression algorithm is not compared. Where DimBridge answers "describe these points", this work additionally asks "find regions worth describing" and "describe them in a way that does not hinge on the exact points".

7.3. Limitations

The predicates are axis-parallel only and so, as noted under Trade-Offs, describe correlated or oblique structure coarsely at best. The dimensionality reductions used in the hierarchy (UMAP, t-SNE) are stochastic algorithms. The experiments of Chapter 6 ran them unseeded. EQ2–EQ4 and the benchmark walk report over repeated builds, whereas the tuning study of Section 6.5 is a single study per grid cell and the PCA and t-SNE arms are single effective draws. Only afterwards, for the case study, did the prototype pin their random state (Section 6.10), so a given configuration now rebuilds identically on the same machine and environment. Repeatable is not robust, however: what the hierarchy finds can still change with the arbitrary seed, as the digits seed sweep at depth one shows. The faithfulness measures quantify how well a projection preserves neighborhoods and distances, which is not the same as whether a region is *semantically* useful to an analyst. A faithful view of uninteresting structure still scores well. Real datasets rarely come with ground-truth subspaces, so the benchmark results are descriptive rather than recovery-based. The per-node embeddings and the $O(n^2)$ faithfulness measures scale poorly to very large nodes, bounding the practical dataset size. And while ten datasets are wired through the registry, only five are exercised by the quantitative experiments (six counting the case-study screening sweep), so the high-dimensional entries go untested.

A structural bound follows from where the clustering happens. The UMAP pre-reduction, when active, is fit once on the root (Section 4.4), so the recursion partitions slices of a single global embedding and cannot recover structure that embedding has already collapsed; the local re-projections adapt the *view* per region, not the space being clustered. Where the pre-reduction is inactive, clustering runs in the full feature space, but the screening sweep shows that variant reaching depth two on only two of the four real datasets (Table 6.8). The configuration that recurses most readily is thus the one whose splits are least anchored in the original features, a trade-off the two evaluated variants sit on opposite sides of (Sections 6.6 and 6.10).

A further limitation is an implementation gap rather than a fundamental one: no automated configuration search is wired into the live tree, so the tuning study of Section 6.5 runs offline. Closing this loop is future work (Chapter 8).

7.4. Threats to Validity

The conclusions are bounded along the three standard axes. The chief *internal* threat is that an observed effect could be an artifact of a particular dataset or a single lucky seed rather than a property of the method. The stochastic, unseeded embeddings make this acute. The mitigations applied were repeated runs over multiple replicates (Section 6.3, except the tuning study) and synthetic data whose ground truth is known. Regions within one build share that build's embedding, so the per-region tests overstate independence. The per-dataset aggregation and the gt_class control (assessed at the class level, Section 6.6) bound, but do not remove, this dependence. The chief *external* threat is generalization: the evaluation spans only a handful of datasets, so findings may not carry to data of very different shape or dimensionality. A second external bound is comparative: every baseline in Chapter 6 is a flat or strict variant of this pipeline, and no system from the strands of Chapter 3 was implemented or run, so Table 3.1 positions the approach by property coverage, not by measured performance. The chief *construct* threat is whether the chosen measures capture the concepts they stand for: whether trustworthiness, continuity, and predicate F1 measure "interesting structure" and "interpretable, stable description", or only convenient proxies for them. One construct caveat is asymmetric across DR methods: stress is, up to normalization, the quantity metric MDS minimizes on the points it is fitted to, and is closely coupled to PCA's reconstruction objective, so the stress half of the H1a result for those two methods largely restates the estimators' fitting criteria. The neighborhood-preservation half, which neither PCA nor MDS optimizes, carries the independent evidence. And any use of class labels carries the "classes are not clusters" caveat [Jeo+24]: agreement with labels is corroboration, not the definition of success.

8. Conclusion

8.1. Summary

High-dimensional data is explored through low-dimensional projections, but a single global projection both hides structure that is local to part of the data and fails to explain, in terms of the original features, why a visible group is a group. These are the two difficulties this thesis set out to address.

In response, the thesis developed a hierarchical, density-based approach: the dataset is recursively decomposed with HDBSCAN, with an optional UMAP pre-reduction and a local projection and faithfulness score computed at each level, so that an analyst can drill from a global overview down to compact, density-defined regions. A selected set of points within a region is then described by axis-parallel range predicates, in both a strict, DimBridge-style form and a relaxed form that trims each feature range inward by a severity-proportional amount to trade coverage for stability. The approach is realized in the HiLDE prototype, and an evaluation protocol combining projection-faithfulness and predicate-level measures was designed around it.

All four evaluation questions were answered by completed experiments, complemented by the benchmark characterization of Section 6.9 and the two walk-throughs of Section 6.10. The verdicts are stated question by question in the next section. Two of the benchmark walk's four pre-specified checks failed, informatively: the local-view advantage proved more uniform across leaves than the earlier paired experiment suggested, and recursion depth compounds the dropped-noise fraction well beyond the depth-two ranges. The expert evidence is two moderated sessions, reported within the bounds stated in Section 6.10.

8.2. Answers to the Evaluation Questions

EQ1 (configuration). The density-clustering stage is far more configuration-sensitive than the two-dimensional view: tuning lifts the in-sample density-validity objective substantially over the shipped defaults (partly off degenerate defaults scored at the floor) while faithfulness sits near its ceiling untuned, and intelligent search pays only on the harder objective (Section 6.5, Table 6.2, Figure 6.2). The protocol check embedded in the question is a partial pass (Verdict, Section 6.5): the implemented measures do discriminate configurations, but density validity and label agreement must be treated as separate axes rather than conflated ($r \approx 0.17$). **EQ2 (local views).** Split verdict, matching its two hypotheses (Section 6.6, Table 6.3): a region's local view is more faithful than a single global projection for PCA and MDS (and marginally UMAP) but not for t-SNE, with a method-neutral control confirming the direction on the three class-bearing real datasets (on the two-dimensional rings the control effect is nil) and significance where enough class regions exist to test. Partitionally the hierarchy over-segments and does *not* recover ground-truth classes better than a flat clustering. **EQ3 (nested structure).** Yes where the predicted mechanism applies (Section 6.7, Table 6.5): the recursion recovers planted nested-subspace structure that an unaided flat clustering merges, at every scale separation, and the advantage reverses on the non-nested control. Together, EQ2 and EQ3 specify the condition: the decomposition pays off when interesting structure nests at several scales in differing subspaces. It is not a method for recovering a dataset's class partition. **EQ4 (relaxation).** No under the pre-specified primary analysis (Section 6.8, Table 6.6, Figures 6.6–6.8): relaxed predicates are more *consistent* (on the real dataset, the F1 standard deviation halves under selection perturbation) but, for the dense predicate, not usefully more robust. The per-clause trim compounds to $\approx t^{d_{\text{eff}}}$ across the conjunction and destroys recall in high dimensions. A favorable operating point does exist for sparse predicates on data with cluster margins ($t^* = 0.95$–0.9), where relaxation also shortens descriptions at negligible cost on wine and digits (median F1 $0.77 \to 0.74$ and $0.94 \to 0.89$), but at a material cost on breast cancer ($0.91 \to 0.78$, Section 6.9). The severity-proportional tail split is not supported, and a symmetric split is the recommended default. This result motivates the per-predicate trim budget described in the next section.

8.3. Future Work

The most pressing items follow directly from the completed experiments, in priority order. First, the per-predicate trim budget the stability experiment motivates: re-parameterize relaxation as $t_{\text{clause}} = t^{1/d_{\text{eff}}}$ (or relax only selected clauses) so that the threshold again means what the design intended, and re-run the stability sweep under it with the ready-made harness. Second, a feature-weighting node splitter that closes the growing margin to the subspace-aware oracle in the planted-subspace experiment. Third, an automated configuration search wired into the live tree, feeding the offline tuning study's findings back into the interactive defaults. Fourth, richer relaxation operators, including an explicit clause-dropping operator and oblique, correlation-aware predicates that move beyond axis-parallel ranges. Fifth, a noise-aware recursion:

points dropped as HDBSCAN noise mid-recursion never reach a leaf and the loss compounds with depth (Section 6.9), so a stopping or reassignment rule that bounds cumulative attrition would directly improve coverage under deep recursion. Sixth, adapting the view layer where EQ2 found it failing: per-node t-SNE parameters, or omitting the local t-SNE re-embedding on small regions, would address the small-region reversal of Section 6.6. Seventh, extending the completed experiments to the registry's remaining, higher-dimensional benchmarks (Olivetti faces, QM9, MNIST/Fashion-MNIST), which no experiment currently covers. Eighth, the remaining qualitative gap (Section 6.10): further expert sessions, and ultimately a user study to validate the interpretability claims. And ninth, scalability improvements to the per-node embedding and faithfulness computation so that the overview remains responsive on very large nodes, where an optional density overlay could also relieve overplotting.

A. Project Resources

The HiLDE web application and its source code are available online:

Website: `https://hilde.3m0.de`

Source code: `https://github.com/hannesmoehring/HiLDE`

Introduction video: `https://youtu.be/BHsBJ5zTIYA`

Abbreviations

EQ Evaluation Question
HiLDE Hierarchical Local Decomposition and Explanation
DR dimensionality reduction
PCA principal component analysis
MDS multidimensional scaling
t-SNE t-distributed stochastic neighbor embedding
UMAP Uniform Manifold Approximation and Projection
DBSCAN Density-Based Spatial Clustering of Applications with Noise
HDBSCAN Hierarchical Density-Based Spatial Clustering of Applications with Noise
GLOSH Global-Local Outlier Score from Hierarchies
RCM retained-center-mass
MRRE mean relative rank errors
CADI class angular distortion index
TPE Tree-structured Parzen Estimator
DBCV density-based clustering validation
ARI adjusted Rand index
NMI normalized mutual information
T&C trustworthiness and continuity

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Bibliography

[Ach+06] E. Achtert, C. Böhm, H.-P. Kriegel, P. Kröger, I. Müller-Gorman, and A. Zimek. "Finding hierarchies of subspace clusters." In: *Knowledge Discovery in Databases: PKDD 2006*. Vol. 4213. Lecture Notes in Computer Science. Springer, 2006, pp. 446–453. DOI: 10.1007/11871637_42.

[Ach+07] E. Achtert, C. Böhm, H.-P. Kriegel, P. Kröger, I. Müller-Gorman, and A. Zimek. "Detection and visualization of subspace cluster hierarchies." In: *Advances in Databases: Concepts, Systems and Applications (DASFAA 2007)*. Vol. 4443. Lecture Notes in Computer Science. Springer, 2007, pp. 152–163. DOI: 10.1007/978-3-540-71703-4_15.

[Agg+99] C. C. Aggarwal, J. L. Wolf, P. S. Yu, C. Procopiuc, and J. S. Park. "Fast algorithms for projected clustering." In: *Proceedings of the 1999 ACM SIGMOD International Conference on Management of Data*. ACM, 1999, pp. 61–72. DOI: 10.1145/304182.304188.

[Agr+98] R. Agrawal, J. Gehrke, D. Gunopulos, and P. Raghavan. "Automatic Subspace Clustering of High Dimensional Data for Data Mining Applications." In: *Proceedings of the 1998 ACM SIGMOD International Conference on Management of Data*. ACM, 1998, pp. 94–105. DOI: 10.1145/276305.276314.

[AHK01] C. C. Aggarwal, A. Hinneburg, and D. A. Keim. "On the surprising behavior of distance metrics in high dimensional space." In: *Database Theory — ICDT 2001*. Springer, 2001, pp. 420–434.

[AJ18] D. Alvarez-Melis and T. S. Jaakkola. *On the Robustness of Interpretability Methods*. Presented at the ICML 2018 Workshop on Human Interpretability in Machine Learning. 2018. DOI: 10.48550/arXiv.1806.08049.

[AK98] E. Alpaydin and C. Kaynak. *Optical Recognition of Handwritten Digits*. UCI Machine Learning Repository. 1998. DOI: 10.24432/C50P49.

[Aki+19] T. Akiba, S. Sano, T. Yanase, T. Ohta, and M. Koyama. "Optuna: A next-generation hyperparameter optimization framework." In: *Proceedings of the 25th ACM SIGKDD international conference on knowledge discovery & data mining*. 2019, pp. 2623–2631.

[Aup07] M. Aupetit. “Visualizing distortions and recovering topology in continuous projection techniques.” In: *Neurocomputing* 70.7-9 (2007), pp. 1304–1330.

[BDF23] A. Bibal, V. Delchevalerie, and B. Frénay. “DT-SNE: t-SNE discrete visualizations as decision tree structures.” In: *Neurocomputing* 529 (2023), pp. 101–112. DOI: `10.1016/j.neucom.2023.01.073`.

[Ber+11] J. Bergstra, R. Bardenet, Y. Bengio, and B. Kégl. “Algorithms for hyper-parameter optimization.” In: *Advances in Neural Information Processing Systems*. Vol. 24. 2011.

[Bey+99] K. Beyer, J. Goldstein, R. Ramakrishnan, and U. Shaft. “When is “nearest neighbor” meaningful?” In: *Database Theory — ICDT’99*. Springer, 1999, pp. 217–235.

[BG05] I. Borg and P. J. F. Groenen. *Modern Multidimensional Scaling: Theory and Applications*. 2nd ed. Springer, 2005.

[Cam+15] R. J. G. B. Campello, D. Moulavi, A. Zimek, and J. Sander. “Hierarchical Density Estimates for Data Clustering, Visualization, and Outlier Detection.” en. In: *ACM Transactions on Knowledge Discovery from Data* 10.1 (July 2015), pp. 1–51. ISSN: 1556-4681, 1556-472X. DOI: `10.1145/2733381`.

[Chu+20] Y. Chung, T. Kraska, N. Polyzotis, K. H. Tae, and S. E. Whang. “Automated data slicing for model validation: A big data-ai integration approach.” In: *IEEE Transactions on Knowledge and Data Engineering* 32.12 (2020), pp. 2284–2296. DOI: `10.1109/TKDE.2019.2916074`.

[CMS13] R. J. G. B. Campello, D. Moulavi, and J. Sander. “Density-Based Clustering Based on Hierarchical Density Estimates.” In: *Advances in Knowledge Discovery and Data Mining*. Ed. by J. Pei, V. S. Tseng, L. Cao, H. Motoda, and G. Xu. Vol. 7819. Series Title: Lecture Notes in Computer Science. Berlin, Heidelberg: Springer Berlin Heidelberg, 2013, pp. 160–172. ISBN: 978-3-642-37456-2. DOI: `10.1007/978-3-642-37456-2_14`.

[Cor+09] P. Cortez, A. Cerdeira, F. Almeida, T. Matos, and J. Reis. “Modeling wine preferences by data mining from physicochemical properties.” In: *Decision Support Systems* 47.4 (2009), pp. 547–553. DOI: `10.1016/j.dss.2009.05.016`.

[Esp+21] M. Espadoto, R. M. Martins, A. Kerren, N. S. T. Hirata, and A. C. Telea. “Toward a Quantitative Survey of Dimension Reduction Techniques.” In: *IEEE Transactions on Visualization and Computer Graphics* 27.3 (Mar. 2021), pp. 2153–2173. ISSN: 1077-2626, 1941-0506, 2160-9306. DOI: 10.1109/TVCG.2019.2944182.

[Est+96] M. Ester, H.-P. Kriegel, J. Sander, and X. Xu. "A density-based algorithm for discovering clusters in large spatial databases with noise." In: *Proceedings of the Second International Conference on Knowledge Discovery and Data Mining (KDD-96)*. Portland, Oregon: AAAI Press, 1996, pp. 226–231.

[FKM20] T. Fujiwara, O.-H. Kwon, and K.-L. Ma. "Supporting Analysis of Dimensionality Reduction Results with Contrastive Learning." In: *IEEE Transactions on Visualization and Computer Graphics* 26.1 (2020), pp. 45–55. DOI: 10.1109/TVCG.2019.2934251.

[GKM26] K. Gunaratne, S. Kobourov, and J. Miller. "Class Angular Distortion Index for Dimensionality Reduction." In: *Computer Graphics Forum*, e70465 (2026). Early View. DOI: 10.1111/cgf.70465.

[HA85] L. Hubert and P. Arabie. "Comparing partitions." In: *Journal of Classification* 2.1 (1985), pp. 193–218. DOI: 10.1007/BF01908075.

[Han+17] M. F. Hanafi, A. Abouzied, L. Chiticariu, and Y. Li. "Seer: Auto-generating information extraction rules from user-specified examples." In: *Proceedings of the 2017 CHI Conference on Human Factors in Computing Systems*. 2017, pp. 6672–6682.

[Her+11] F. Herrera, C. J. Carmona, P. González, and M. J. Del Jesus. "An overview on subgroup discovery: foundations and applications." en. In: *Knowledge and Information Systems* 29.3 (Dec. 2011), pp. 495–525. ISSN: 0219-1377, 0219-3116. DOI: 10.1007/s10115-010-0356-2.

[Hol79] S. Holm. "A simple sequentially rejective multiple test procedure." In: *Scandinavian Journal of Statistics* 6.2 (1979), pp. 65–70.

[Hub85] P. J. Huber. "Projection pursuit." In: *The Annals of Statistics* 13.2 (1985), pp. 435–475.

[Hun+16] M. Hund, D. Böhm, W. Sturm, M. Sedlmair, T. Schreck, T. Ullrich, D. A. Keim, L. Majnaric, and A. Holzinger. "Visual analytics for concept exploration in subspaces of patient groups: Making sense of complex datasets with the Doctor-in-the-loop." In: *Brain Informatics* 3.4 (2016), pp. 233–247. DOI: 10.1007/s40708-016-0043-5.

[Jeo+22] H. Jeon, H.-K. Ko, J. Jo, Y. Kim, and J. Seo. "Measuring and explaining the inter-cluster reliability of multidimensional projections." In: *IEEE Transactions on Visualization and Computer Graphics* 28.1 (2022), pp. 551–561. DOI: 10.1109/TVCG.2021.3114833.

[Jeo+23] H. Jeon, A. Cho, J. Jang, S. Lee, J. Hyun, H.-K. Ko, J. Jo, and J. Seo. "ZADU: A Python Library for Evaluating the Reliability of Dimensionality Reduction Embeddings." In: *2023 IEEE Visualization and Visual Analytics (VIS)*. Melbourne, Australia: IEEE, Oct. 2023, pp. 196–200. ISBN: 979-8-3503-2557-7. DOI: 10.1109/VIS54172.2023.00048.

[Jeo+24] H. Jeon, Y.-H. Kuo, M. Aupetit, K.-L. Ma, and J. Seo. "Classes are not clusters: Improving label-based evaluation of dimensionality reduction." In: *IEEE Transactions on Visualization and Computer Graphics* 30.1 (2024), pp. 781–791. DOI: 10.1109/TVCG.2023.3327187.

[KKK04] K. Kailing, H.-P. Kriegel, and P. Kröger. "Density-connected subspace clustering for high-dimensional data." In: *Proceedings of the 2004 SIAM International Conference on Data Mining*. SIAM, 2004, pp. 246–256. DOI: 10.1137/1.9781611972740.23.

[KKZ09] H.-P. Kriegel, P. Kröger, and A. Zimek. "Clustering High-Dimensional Data: A Survey on Subspace Clustering, Pattern-Based Clustering, and Correlation Clustering." In: *ACM Transactions on Knowledge Discovery from Data* 3.1 (2009), pp. 1–58. DOI: 10.1145/1497577.1497578.

[KMB12] F. Keller, E. Müller, and K. Böhm. "HiCS: High contrast subspaces for density-based outlier ranking." In: *2012 IEEE 28th International Conference on Data Engineering*. IEEE, 2012, pp. 1037–1048. DOI: 10.1109/ICDE.2012.88.

[Kru64] J. B. Kruskal. "Multidimensional scaling by optimizing goodness of fit to a nonmetric hypothesis." In: *Psychometrika* 29.1 (1964), pp. 1–27.

[Lai+25] F. Lai, E. Heiter, G. Bied, and J. Lijffijt. "InfoClus: Informative clustering of high-dimensional data embeddings." In: *Machine Learning and Knowledge Discovery in Databases (ECML PKDD 2025), Research Track*. Vol. 16013. Lecture Notes in Computer Science. Springer, 2025, pp. 480–497. DOI: 10.1007/978-3-032-05962-8_28.

[Lee77] J. de Leeuw. "Applications of convex analysis to multidimensional scaling." In: *Recent Developments in Statistics*. Ed. by J.-R. Barra, F. Brodeau, G. Romier, and B. Van Cutsem. North-Holland, 1977, pp. 133–145.

[LFK08] D. Leman, A. Feelders, and A. Knobbe. "Exceptional model mining." In: *Joint European conference on machine learning and knowledge discovery in databases*. Springer, 2008, pp. 1–16.

[Liu+15] S. Liu, B. Wang, J. J. Thiagarajan, P.-T. Bremer, and V. Pascucci. "Visual Exploration of High-Dimensional Data through Subspace Analysis and Dynamic Projections." In: *Computer Graphics Forum* 34.3 (2015), pp. 271–280. DOI: 10.1111/cgf.12639.

[Liu+17] S. Liu, D. Maljovec, B. Wang, P.-T. Bremer, and V. Pascucci. "Visualizing high-dimensional data: Advances in the past decade." In: *IEEE Transactions on Visualization and Computer Graphics* 23.3 (2017), pp. 1249–1268. DOI: 10.1109/TVCG.2016.2640960.

[LV09] J. A. Lee and M. Verleysen. "Quality assessment of dimensionality reduction: Rank-based criteria." In: *Neurocomputing* 72.7-9 (2009), pp. 1431–1443.

[Mar+21] W. E. Marcílio-Jr, D. M. Eler, F. V. Paulovich, J. F. Rodrigues-Jr, and A. O. Artero. "ExplorerTree: A focus+context exploration approach for 2D embeddings." In: *Big Data Research* 25 (2021), p. 100239. DOI: 10.1016/j.bdr.2021.100239.

[Mar+25] W. E. Marcílio-Jr, D. M. Eler, F. V. Paulovich, and R. M. Martins. "HUMAP: Hierarchical uniform manifold approximation and projection." In: *IEEE Transactions on Visualization and Computer Graphics* 31.9 (2025), pp. 5741–5753. DOI: 10.1109/TVCG.2024.3471181.

[MH17] L. McInnes and J. Healy. "Accelerated Hierarchical Density Based Clustering." In: *2017 IEEE International Conference on Data Mining Workshops (ICDMW)*. New Orleans, LA: IEEE, Nov. 2017, pp. 33–42. ISBN: 978-1-5386-3800-2. DOI: 10.1109/ICDMW.2017.12.

[MHA17] L. McInnes, J. Healy, and S. Astels. "hdbscan: Hierarchical density based clustering." In: *The Journal of Open Source Software* 2.11 (Mar. 2017), p. 205. ISSN: 2475-9066. DOI: 10.21105/joss.00205.

[MHM18a] L. McInnes, J. Healy, and J. Melville. *UMAP: Uniform manifold approximation and projection for dimension reduction*. 2018. DOI: 10.48550/arXiv.1802.03426.

[MHM18b] L. McInnes, J. Healy, and J. Melville. *Using UMAP for Clustering*. umap-learn documentation. 2018. URL: https://umap-learn.readthedocs.io/en/latest/clustering.html (visited on 08/16/2026).

[Mon+22] B. Montambault, C. D. Brumar, M. Behrisch, and R. Chang. *PIXAL: Anomaly reasoning with visual analytics*. 2022. DOI: 10.48550/arXiv.2205.11004.

[Mon+25] B. Montambault, G. Appleby, J. Rogers, C. D. Brumar, M. Li, and R. Chang. "DimBridge: Interactive explanation of visual patterns in dimensionality reductions with predicate logic." In: *IEEE Transactions on Visualization and Computer Graphics* 31.1 (2025), pp. 207–217. DOI: 10.1109/TVCG.2024.3456391.

[Mou+14] D. Moulavi, P. A. Jaskowiak, R. J. G. B. Campello, A. Zimek, and J. Sander. "Density-Based Clustering Validation." en. In: *Proceedings of the 2014 SIAM International Conference on Data Mining*. Society for Industrial and Applied Mathematics, Apr. 2014, pp. 839–847. ISBN: 978-1-61197-344-0. DOI: 10.1137/1.9781611973440.96.

[Mül+09] E. Müller, I. Assent, S. Günnemann, R. Krieger, and T. Seidl. "Relevant subspace clustering: Mining the most interesting non-redundant concepts in high dimensional data." In: *2009 Ninth IEEE International Conference on Data Mining*. IEEE, 2009, pp. 377–386.

[Mun09] T. Munzner. "A Nested Model for Visualization Design and Validation." In: *IEEE Transactions on Visualization and Computer Graphics* 15.6 (2009), pp. 921–928. DOI: 10.1109/TVCG.2009.111.

[NA19] L. G. Nonato and M. Aupetit. "Multidimensional projection for visual analytics: Linking techniques with distortions, tasks, and layout enrichment." In: *IEEE Transactions on Visualization and Computer Graphics* 25.8 (2019), pp. 2650–2673. DOI: 10.1109/TVCG.2018.2846735.

[Ped+11] F. Pedregosa, G. Varoquaux, A. Gramfort, V. Michel, B. Thirion, O. Grisel, M. Blondel, P. Prettenhofer, R. Weiss, V. Dubourg, J. Vanderplas, A. Passos, D. Cournapeau, M. Brucher, M. Perrot, and É. Duchesnay. "Scikit-learn: Machine learning in Python." In: *Journal of Machine Learning Research* 12 (2011), pp. 2825–2830.

[Pez+16] N. Pezzotti, T. Höllt, B. P. F. Lelieveldt, E. Eisemann, and A. Vilanova. "Hierarchical Stochastic Neighbor Embedding." In: *Computer Graphics Forum* 35.3 (2016), pp. 21–30. DOI: 10.1111/cgf.12878.

[RH07] A. Rosenberg and J. Hirschberg. "V-measure: A conditional entropy-based external cluster evaluation measure." In: *Proceedings of the 2007 Joint Conference on Empirical Methods in Natural Language Processing and Computational Natural Language Learning (EMNLP-CoNLL)*. 2007, pp. 410–420.

[RSG18] M. T. Ribeiro, S. Singh, and C. Guestrin. "Anchors: High-precision model-agnostic explanations." In: *Proceedings of the AAAI Conference on Artificial Intelligence*. Vol. 32. 1. 2018. DOI: 10.1609/aaai.v32i1.11491.

[Sac+17] D. Sacha, L. Zhang, M. Sedlmair, J. A. Lee, J. Peltonen, D. Weiskopf, S. C. North, and D. A. Keim. "Visual interaction with dimensionality reduction: A structured literature analysis." In: *IEEE Transactions on Visualization and Computer Graphics* 23.1 (2017), pp. 241–250. DOI: 10.1109/TVCG.2016.2598495.

[Sal+24] P. Salmanian, A. Chatzimparmpas, A. C. Karaca, and R. M. Martins. "DimVis: Interpreting Visual Clusters in Dimensionality Reduction With Explainable Boosting Machine." In: *Machine Learning Methods in Visualisation for Big Data (MLVis)*. The Eurographics Association, 2024. DOI: 10.2312/mlvis.20241125.

[Sil+15] R. R. O. da Silva, P. E. Rauber, R. M. Martins, R. Minghim, and A. C. Telea. "Attribute-based Visual Explanation of Multidimensional Projections." In: *EuroVis Workshop on Visual Analytics (EuroVA)*. The Eurographics Association, 2015, pp. 31–35. DOI: 10.2312/eurova.20151100.

[SMM12] M. Sedlmair, M. Meyer, and T. Munzner. "Design Study Methodology: Reflections from the Trenches and the Stacks." In: *IEEE Transactions on Visualization and Computer Graphics* 18.12 (2012), pp. 2431–2440. DOI: 10.1109/TVCG.2012.213.

[SMT13] M. Sedlmair, T. Munzner, and M. Tory. "Empirical guidance on scatterplot and dimension reduction technique choices." In: *IEEE Transactions on Visualization and Computer Graphics* 19.12 (2013), pp. 2634–2643.

[Tor52] W. S. Torgerson. "Multidimensional scaling: I. Theory and method." In: *Psychometrika* 17.4 (1952), pp. 401–419. DOI: 10.1007/BF02288916.

[Une+17] V. van Unen, T. Höllt, N. Pezzotti, N. Li, M. J. T. Reinders, E. Eisemann, F. Koning, A. Vilanova, and B. P. F. Lelieveldt. "Visual analysis of mass cytometry data by hierarchical stochastic neighbour embedding reveals rare cell types." In: *Nature Communications* 8.1 (2017), p. 1740. DOI: 10.1038/s41467-017-01689-9.

[VH08] L. Van der Maaten and G. Hinton. "Visualizing data using t-SNE." In: *Journal of Machine Learning Research* 9 (2008), pp. 2579–2605.

[VK01] J. Venna and S. Kaski. "Neighborhood preservation in nonlinear projection methods: An experimental study." In: *Artificial Neural Networks — ICANN 2001*. Series Title: Lecture Notes in Computer Science, vol. 2130. Berlin, Heidelberg: Springer Berlin Heidelberg, 2001, pp. 485–491. DOI: 10.1007/3-540-44668-0_68.

[Wil45] F. Wilcoxon. "Individual comparisons by ranking methods." In: *Biometrics Bulletin* 1.6 (1945), pp. 80–83. DOI: 10.2307/3001968.

[WM18] B. Wang and K. Mueller. "The Subspace Voyager: Exploring high-dimensional data along a continuum of salient 3D subspaces." In: *IEEE Transactions on Visualization and Computer Graphics* 24.2 (2018), pp. 1204–1222. DOI: 10.1109/TVCG.2017.2672987.

[WVJ16] M. Wattenberg, F. Viégas, and I. Johnson. "How to Use t-SNE Effectively." In: *Distill* (2016). DOI: 10.23915/distill.00002.